

Supplementary Material: Discovery of a Conserved Bulged-Rod RNA Envelope in Obelisks Supporting a Testable Host-Polymerase Recruitment Hypothesis

Kevin Christopher Lin
Great Neck South High School, Great Neck, NY, USA
kevin.lin.2028@gmail.com

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S1 Cross-class topology comparison

To test for cross-class structural distinctness on the same axes that define the BRRC envelope, fraction paired (FP), small-loop density (SLD), and minimum-free-energy per nucleotide (MFE/nt) in each class were measured (Table S1). The panel includes obelisk cluster representatives ($n=27$), HDV-like genomes ($n=25$), and 24 pooled Rfam families ($n=750$). Notably, fraction paired and small-loop density separated obelisks and HDV-like genomes from the Rfam

panel with Cliff's $\delta = +0.91$. This is a very large effect size by the conventional Cliff thresholds of $|\delta| > 0.474$. In contrast, MFE per nucleotide did not separate the classes. This is consistent with a topology-driven rather than a thermodynamic-driven signature. Together, these results show that the BRRC axes capture a structural class signature, suggesting that obelisks and HDV-like genomes share a topological feature that does not reduce to thermodynamic stability.

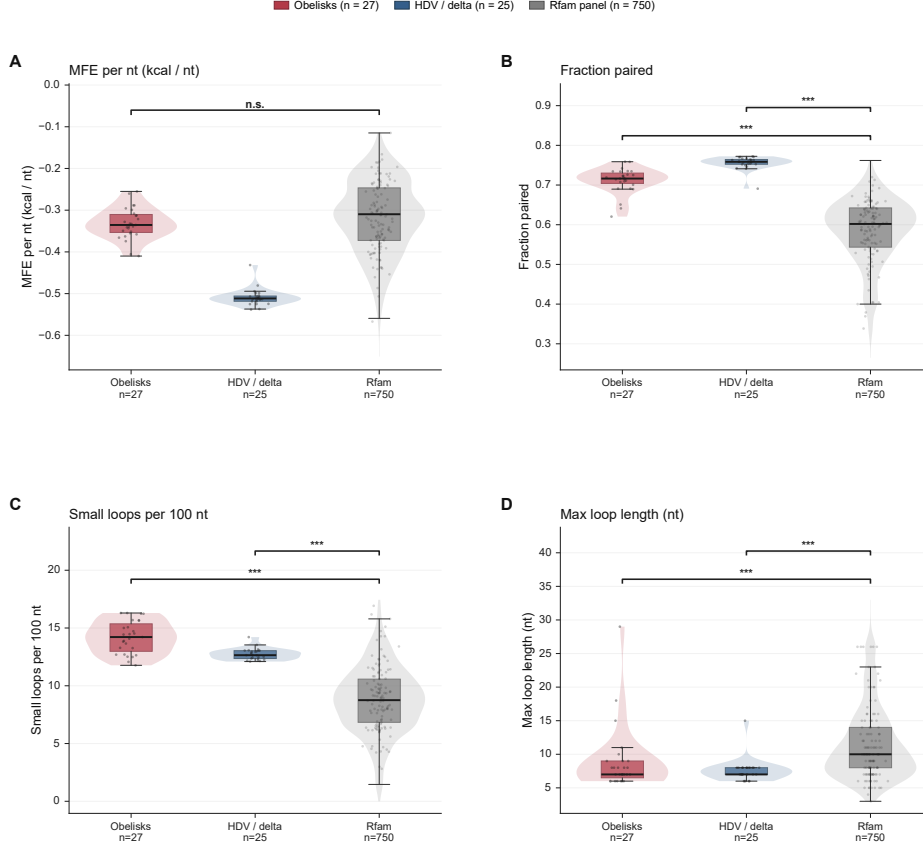


Figure S1: Topology separates obelisks from generic structured RNAs across four metrics. Box plots with individual sequences overlaid include obelisk cluster representatives ($n=27$, $\geq 95\%$ -identity reps from the Zheludev catalog), HDV-like complete genomes ($n=25$ representative species), and pooled Rfam families ($n=750$). Fraction paired and small-loop density separate obelisks and HDV-like from Rfam with very large effect sizes, while MFE/nt does not. The HDV-like panel here is representative-species, while the full-class pass-rate panel (Figure S5A) uses all $n=47$ delta and delta-like genomes.

Table S2: Per-family means in the expanded Rfam panel ($n=750$ total; $n=30$ per family, except $n=60$ for RF00059 TPP). SLD = small-loop density per 100 nt. The length-window-restricted bacterial-RNA comparison in Section 3.5 uses a stricter subset matching the obelisk window (700–1,700 nt): $n=2$ RF00010 (RNase P), $n=28$ RF00028 (group I introns), $n=266$ RF00177 (bacterial small SRP) sequences pass the length filter.

Rfam ID	Family	n	MFE/nt	Frac. paired	SLD	Max loop
RF00010	RNase P type A	30	-0.418	0.630	9.07	11.8
RF00011	RNase P type B	30	-0.307	0.589	8.40	15.9
RF00012	U3 snoRNA	30	-0.298	0.650	9.17	10.3
RF00013	6S RNA	30	-0.381	0.668	10.51	11.2
RF00023	tmRNA	30	-0.366	0.616	7.85	13.9

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Rfam ID	Family	n	MFE/nt	Frac. paired	SLD	Max loop
RF00028	Group I intron	30	-0.258	0.604	8.72	14.3
RF00029	Group II intron	30	-0.357	0.664	10.23	7.9
RF00031	SECIS	30	-0.297	0.617	8.33	7.2
RF00033	MicF sRNA	30	-0.213	0.404	6.28	25.3
RF00040	rne5 sRNA	30	-0.306	0.621	8.98	11.3
RF00050	FMN riboswitch	30	-0.379	0.591	10.56	7.4
RF00057	RyhB sRNA	30	-0.242	0.435	3.22	12.7
RF00059	TPP riboswitch	60	-0.317	0.573	9.25	9.2
RF00099	snoR9	30	-0.354	0.623	10.25	9.2
RF00162	SAM riboswitch	30	-0.333	0.617	9.20	8.4
RF00174	Cobalamin riboswitch	30	-0.378	0.607	9.57	8.8
RF00177	Bacterial small SRP	30	-0.341	0.629	9.86	12.1
RF00229	PSTVd Avsunviroidae	30	-0.321	0.635	8.51	16.4
RF00379	Ribosomal T-box	30	-0.367	0.601	9.63	9.6
RF00543	PSTVd Pospiviroidae	30	-0.227	0.528	6.95	14.4
RF01051	c-di-GMP-I riboswitch	30	-0.299	0.550	11.22	8.6
RF01054	preQ1-II riboswitch	30	-0.215	0.513	5.46	14.3
RF01057	SAH riboswitch	30	-0.369	0.600	9.32	8.0
RF01998	OxyS sRNA	30	-0.245	0.511	8.67	9.6

S2 Threshold sensitivity sweeps

The BRRC envelope is stable across the choice of cutoff. Obelisks remained enriched over dinucleotide shuffles across a wide range around each threshold (Tables S3-S5). The chosen operating points were picked to balance obelisk retention against shuffle rejection rather than to maximize enrichment. Importantly, the sweeps were computed on the full Zheludev catalog ($n=5,169$ obelisks) against length-matched dinucleotide shuffles, indicating that the threshold robustness applies at the same catalog scale as the headline analyses.

S3 High-order null models for the BRRC signal

To test whether composition at orders higher than dinucleotide explained the BRRC signal, the joint pass rate in each null was measured against 3 higher-order null models that progressively preserve more sequence structure (Table S6). Trinucleotide-Eulerian shuffles passed at 1.34%. Per-sequence Markov-3 resamples passed at 2.11%. Synonymous codon-frequency-preserving permutations of the Oblin-1 ORF passed at 16.6%. In contrast, the unshuffled obelisks passed at 89.65%. All values are at full-catalog scale ($n_{\text{src}}=5,169$, $n_{\text{shuf}}=15,507$ each, $n_{\text{rep}}=3$ replicates per obelisk per shuffle). The BRRC envelope is therefore not explained by triplet, 4-mer, or codon-level composition. For example, the trinucleotide-Eulerian shuffle uses a proper Hierholzer’s algorithm walk on the de Bruijn 2-mer/3-mer graph. The path was verified to consume every edge and to end at the original sequence’s terminal 2-mer. Strikingly, 0 of 15,507 shuffles fell back to the unshuffled source sequence under this implementation (`results/json/trinuc_eulerian.json`), indicating that the higher-order null does not collapse onto the original sequence as a degenerate case.

Table S1: Obelisks vs HDV vs Rfam on four topology metrics: means $\pm$ SD with Mann–Whitney U and Cliff’s δ . Bonferroni-corrected significance at $\alpha = 0.05/12 = 0.00417$. *** $p < 0.001$; n.s. = not significant.

Metric	OBELISK ($n=27$)	DELTA ($n=25$)	RFAM ($n=750$)
MFE/nt (kcal/nt)	-0.332 ± 0.037	-0.509 ± 0.020	-0.316 ± 0.088
Fraction paired	0.711 ± 0.032	0.756 ± 0.016	0.586 ± 0.082
Small-loop density (per 100 nt)	14.20 ± 1.40	12.74 ± 0.49	8.74 ± 2.85
Max loop length (nt)	8.93 ± 4.88	7.56 ± 1.73	11.50 ± 5.35
<i>Mann–Whitney U with Cliff’s δ</i>			
MFE/nt: OBELISK vs RFAM	$p=0.13$, $\delta = -0.19$ (n.s.)		
Fraction paired: OBELISK vs RFAM	$p=1.6 \times 10^{-15}$, $\delta = +0.91$ ***		
Small-loop density: OBELISK vs RFAM	$p=7.3 \times 10^{-16}$, $\delta = +0.91$ ***		
Max loop length: OBELISK vs RFAM	$p=7.1 \times 10^{-4}$, $\delta = -0.39$ ***		

Table S3: Fraction-paired threshold sweep on $n=5,169$ obelisks vs 5,169 dinucleotide shuffles; chosen threshold bolded.

Threshold	OBELISK pass	SHUFFLE pass	Enrichment
> 0.55	100.0%	95.9%	$1.0\times$
> 0.59	99.9%	60.1%	$1.7\times$
> 0.63	98.9%	9.4%	$10.5\times$
> 0.65 (chosen)	95.8%	1.7%	$56\times$
> 0.67	86.7%	0.1%	$1,121\times$
> 0.70	70.4%	0.0%	∞

Table S4: Small-loop-density threshold sweep (per 100 nt); chosen threshold bolded.

Threshold	OBELISK pass	SHUFFLE pass	Enrichment
> 7	100.0%	87.4%	$1.1\times$
> 9	99.8%	23.1%	$4.3\times$
> 10 (chosen)	98.5%	5.1%	$19\times$
> 12	83.8%	0.02%	$4,330\times$
> 14	40.2%	0.0%	∞

S4 Loop-context motif enrichment in obelisks vs HDV+viroids

To test for loop-context sequence preferences that distinguish obelisks from HDV-like genomes and viroids, a loop-context 6-mer scan was run over 200 obelisks (57,836 loop 6-mers) versus 47 HDV-like genomes plus 40 viroids (23,009 loop 6-mers). The scan returned 390/1,077 enriched and 252/1,077 depleted 6-mers at Benjamini–Hochberg false discovery rate $\leq 1\%$. This tracks the GC-content gap between bacterial-host obelisks and HDV/viroids (full per-6-mer table in `results/json/loop_motif_enrichment.json`). Together, these results show that the loop-context 6-mer enrichment pattern is largely explained by the host-kingdom composition gap rather than by a discrete loop-binding motif, indicating that the BRRC envelope captures topology rather than a specific loop-resident sequence signal.

Table S5: Maximum-loop-length threshold sweep ($\leq$); chosen threshold bolded.

Threshold	OBELISK pass	SHUFFLE pass	Enrichment
≤ 10	68.0%	0.9%	73 $\times$
≤ 12	79.7%	5.6%	14 $\times$
≤ 15 (chosen)	90.8%	22.8%	4.0$\times$
≤ 18	95.4%	47.0%	2.0 $\times$
≤ 22	97.7%	74.0%	1.3 $\times$

Table S6: Obelisks vs three high-order nulls on joint circular BRRC pass rate, all at full-catalog scale ($n_{\text{src}}=5,169$, $n_{\text{shuf}}=15,507$ each, $n_{\text{rep}}=3$ replicates per obelisk per shuffle). Trinucleotide-Eulerian preserves all 64 triplet counts and the terminal 2-mers (Hierholzer walk on 2-mer nodes / 3-mer edges; `results/json/trinuc_eulerian.json`). Markov-3 fits $P(b_i \mid b_{i-3}b_{i-2}b_{i-1})$ per sequence and resamples length-matched, preserving all 4-mer transition probabilities (`results/json/higher_order_nulls.json`). Synonymous codon-frequency-preserving permutes the positional order of codons within each amino-acid class along the Oblin-1 ORF, preserving the encoded protein, the empirical codon frequencies, and the untranslated regions; UTRs are unchanged. Wilson 95% CIs on each proportion.

Sequence set	Pass rate (%)	Wilson 95% CI	Enrichment ($\times$)
Obelisk catalog ($n=5,169$, full)	89.65	[88.8, 90.4]	1.00
Trinucleotide-Eulerian shuffled (full, $n_{\text{shuf}}=15,507$)	1.341	[1.17, 1.54]	66.8 $\times$ [58, 77]
Markov-3 resampled (full, $n_{\text{shuf}}=15,507$)	2.11	[1.89, 2.34]	42.5 $\times$ [38, 47]
Synonymous codon-frequency-preserving (full, $n_{\text{shuf}}=15,507$)	16.64	[16.06, 17.24]	5.39 $\times$ [5.20, 5.59]

S5 Sequence-HMM and structure-search homology negative for Oblin-1

To bound the degree to which Oblin-1 belongs to a known protein family, 4 complementary homology searches were run on the Oblin-1 reference sequence. The searches were sequence based, profile hidden-Markov-model (HMM), HMM-HMM, and structural. All 4 returned zero hits, or only marginal non-significant hits. This is consistent with Oblin-1 being a novel protein family at every currently indexed homology layer. Importantly, the four searches together cover the major homology layers used in modern protein-family annotation, indicating that the homology-negative result is unlikely to be a single-tool artifact.

Table S7: Four complementary homology searches against the Oblin-1 reference protein (obelisk-S.s, 202 aa). All four return zero or only marginal-non-significant hits. HHpred is a profile-profile (HMM-HMM) search whose query MSA was built by HHblits against UniRef30 (3 iterations); the resulting effective-sequence count $N_{\text{eff}}=1$ shows that Oblin-1 has essentially no detectable homologs deep enough in UniRef30 to build a meaningful profile.

Search	Database / coverage	n significant hits
Foldseek 3diaa	PDB100, AFDB50, GMGCL, MGnify-ESM30, CATH50 (best $e=1.4$)	0 at $e \leq 0.05$
InterProScan (EBI REST)	Pfam, SMART, Gene3D, CDD, NCBI-fam, PIRSF, ProSite, PANTHER	0 across all libraries
pyhmmer phmmer	Pfam-A 37.0 (27,481 HMMs, local scan)	0 at $E \leq 1$
HHpred (MPI Toolkit, HHsearch)	PDB_mmCIF70 (75,560 HMMs, HHblits-UniRef30 3-iter MSA, $N_{\text{eff}}=1$, best Prob 15.1%, $E=430$)	0 at Prob $\geq 50\%$ or $E \leq 10^{-3}$

S6 Independent-engine LinearFold cross-check

The BRRC signal is not specific to ViennaRNA’s dynamic-programming recursion or its Turner-2004 thermodynamic parameters. Re-folding through LinearFold’s algorithm and through CONTRAfold’s machine-learning-trained parameters separates the two axes. Both also reproduce the enrichment when compared against engine-matched dinucleotide shuffles. Together, these results show that the headline BRRC envelope is preserved across a complete change of folding engine and energy model, indicating that the signal is unlikely to be a single-engine artifact.

Table S8: Independent-engine BRRC cross-check using LinearFold (Huang et al., 2019) on a 100-obelisk random subsample. LinearFold is a linear-time beam-search algorithm fully independent of ViennaRNA’s $O(N^3)$ recursion and implementation. Two parameter sets were compiled from source: `linearfold_v` uses Vienna 2004 energy parameters (same thermodynamic table as ViennaRNA, fully independent algorithm), and `linearfold_c` uses CONTRAfold machine-learning-trained parameters (fully parameter-independent). The 97% per-obelisk concordance between ViennaRNA and LinearFold-Vienna establishes that the BRRC signal is not specific to ViennaRNA’s algorithm. Under CONTRAfold parameters, native obelisks pass at 39% (Wilson 95% CI [30.0%, 48.8%]) against 0% on matched CONTRAfold-folded dinucleotide shuffles (0/1,000; one-sided 95% upper bound 0.3% by rule of three; $n_{\text{shuf}}=10$ per obelisk, seed 42), giving an engine-internal lower-bound enrichment of $\geq 130\times$ that confirms the structural signal under a complete change of scoring model.

Engine / parameters	BRRC pass (%)	Concordance with ViennaRNA (%)
ViennaRNA 2.7.2 (Turner 2004)	88	—
LinearFold (Vienna 2004 parameters)	87	97
LinearFold (CONTRAfold ML parameters)	39	49
Dinucleotide-shuffled control (Vienna)	0.35	—
Dinucleotide-shuffled control (LinearFold-CONTRAfold, matched)	0 (0/1,000)	—

S7 TM-align cross-predictor agreement on Oblin-1

To test for cross-predictor consistency of the Oblin-1 fold, TM-align comparisons of the N-globule, the full model, and the MSA-free ESMFold baseline were used. These split the AlphaFold3 $\leftrightarrow$ Boltz-2 fold agreement into a piece that comes from the shared multiple-sequence-alignment (MSA) prior and a piece that survives an MSA-free control. The N-globule TM-align overlap is the most stable cross-predictor signal. In contrast, Domain-A geometry remained predictor sensitive, indicating that the global N-globule topology is more robust to predictor choice than the local Domain-A geometry.

Table S9: AlphaFold3 and Boltz-2 agree on the Oblin-1 N-globule rigid core (TM-score 0.89, RMSD 1.11 Å, residues 1–130) and on the full model (TM 0.65, RMSD 1.76 Å); ESMFold disagrees with AF3 (TM 0.23, RMSD 5.11 Å), confirming the single-sequence baseline is uninformative. TM-align (Zhang and Skolnick, 2005) normalizes RMSD by sequence length; conventional cutoffs are TM > 0.5 same fold, > 0.7 high similarity, < 0.17 random.

Comparison	n residues	TM-score	RMSD (Å)
AF3 vs Boltz-2 (full)	202	0.65	1.76
AF3 vs Boltz-2 (N-globule, residues 1–130)	130	0.89	1.11
AF3 vs ESMFold (full)	202	0.23	5.11

S8 CRISPR spacer search against obelisks

To test whether obelisks carry historical evidence of bacterial CRISPR-Cas targeting, the full set of unique CRISPRCasdb spacers was searched against the Zheludev catalog. Strikingly, zero of 14,111 unique CRISPRCasdb (Pourcel et al., 2020) spacers (covering $\sim 2,000$ bacterial species) hit any of the 5,169 Zheludev-catalog obelisks under a relaxed seed-and-extend search (16-mer exact seed, ≤ 4 mismatches across the full spacer span, both strands, circular folding; per-hit JSON in `results/json/crispr_spacer_hits.json`). This indicates that obelisks have either escaped CRISPR surveillance or have not yet been catalogued by host immune systems at the resolution of the current spacer database.

S9 Chi subfamily outlier analysis

The chi subfamily ($n=30$, 0% BRRC pass) failed the envelope through a single coherent failure mode rather than through a sampling artifact. Strikingly, all 30 members converged on the same max-loop length of 29 nt. This pointed to a stable bulged-rod-plus-large-bubble variant, with a striking resemblance to the central single-stranded segment of bacterial 6S RNA (see also Supplement S29 for the 6S architecture comparison). Together, these results show that the chi cluster is a coherent architectural variant rather than a random scattering of envelope failures, suggesting that the BRRC envelope is correctly excluding a structurally distinct subclass rather than misclassifying a heterogeneous mix.

Table S10: All 30 chi-subfamily members (0% BRRC pass) fail by the same single mode: max-loop length of exactly 29 nt, compared with a 4,533-pass-obelisk baseline mean of 8.5 nt. The 30 unique sequences form a coherent subfamily (pairwise 8-mer Jaccard 0.87, vs 0.78 within alpha as a sanity check); length, fraction paired, and small-loop density sit within typical obelisk ranges. The chi architecture is a stable bulged-rod-plus-large-bubble variant, conceptually analogous to bacterial 6S RNA’s central single-stranded segment. Drawn from FASTA cluster annotations in the Zheludev catalog.

Metric	Chi (n=30)	BRRC-pass baseline (n=4,533)
mean length (nt)	1,228	1,068
mean fraction paired	0.636	0.718
mean small-loop density (per 100 nt)	~ 13	13.7
mean max-loop length (nt)	29.0 (all members exactly 29)	8.5
mean MFE per nt	similar	−0.317
pairwise 8-mer Jaccard	0.87	—
BRRC pass	0/30 (0%)	4,533/4,533 (100%)
failure mode	all 30 fail max_loop ≤ 15 criterion	
		—

S10 Expanded bacterial promoter / UP-element scan

To test whether obelisks resemble any major class of bacterial promoter beyond the canonical σ^{70} motif, a 12-class scan was used. The scan extended the negative σ^{70} promoter result to other bacterial σ factors and to the Gram-positive Pribnow and UP-element classes. No class showed enrichment in obelisks. In contrast, the UP-element A/T tract was actively depleted rather than enriched, indicating that obelisks do not carry promoter-like sequence features under any of the major bacterial promoter consensus patterns tested.

Table S11: No canonical bacterial promoter class shows enrichment in obelisks (Katz 95% CI not strictly above 1.0 for any of 12 classes), and the UP-element-like A/T tract is actively depleted (16.6% vs 35.0% in shuffles, $0.47\times$). The scan tabulates the fraction of sequences with at least one match across 12 motif patterns on $n=500$ obelisks and $n=500$ dinucleotide-shuffled controls (both strands, circular wrap-around), broadening the negative-result argument against bacterial promoter mimicry beyond the single σ^{70} test.

Motif class	Pattern	Obelisk % with hit	Shuffle % with hit	Enrichment
$\sigma^{70} -10$	TA [ATCG] [ATCG] [ATCG] T	100.0	100.0	$1.00\times$
$\sigma^{70} -35$	TTGACA	35.8	44.4	$0.81\times$
Extended $\sigma^{70} -10$	TG [ATCG] TATAAT	1.4	2.4	$0.58\times$
$\sigma^{54} -24$	TGGCAC	19.6	15.0	$1.31\times$
$\sigma^{54} -12$	TTGC [AT] [AT]	71.8	76.4	$0.94\times$
$\sigma^{32} -10$	CCCCAT [ATCG] [ATCG] A	2.6	7.0	$0.37\times$
$\sigma^{32} -35$	CTTGAA	68.0	66.2	$1.03\times$
$\sigma^{38} -10$	TA [ATCG] [ATCG] [ATCG] T	100.0	100.0	$1.00\times$
$\sigma^{28} -10$	GCCGATAA	0.8	1.6	$0.50\times$
$\sigma^{28} -35$	TAAA	99.8	100.0	$1.00\times$
UP-element A/T tract	[AT]{12,}	16.6	35.0	$0.47\times$ (depleted)
Pribnow box (Gram ⁺)	TA [ATCG] [ATCG] AT	99.6	99.8	$1.00\times$

S11 Boltzmann-ensemble BRRC pass probability

The BRRC envelope is a property of the thermodynamic ensemble rather than only of the single MFE conformer. Sampling 30 structures per obelisk from the partition function on a 200-obelisk subset gave a typical-obelisk ensemble pass probability of 0.86 (median 0.97), indicating that the BRRC envelope is satisfied across the bulk of the Boltzmann ensemble and not just by the lowest-energy representative.

Table S12: The typical obelisk’s Boltzmann ensemble satisfies BRRC: mean per-obelisk pass probability 0.86 (median 0.97), with 55/200 obelisks passing on all samples and 195/200 passing on at least one. For each obelisk in a 200-obelisk random subsample (seed 42, length window 700–1,700 nt) we drew 30 structures from the circular Boltzmann ensemble via ViennaRNA **pbacktrack** and computed BRRC on each. The envelope is therefore satisfied by the bulk of the thermodynamic ensemble, not only by the MFE structure.

Statistic	Value
n obelisks sampled	200
n Boltzmann samples per obelisk	30
mean per-obelisk ensemble pass probability	0.859
median per-obelisk ensemble pass probability	0.967
25th percentile pass prob.	0.833
75th percentile pass prob.	1.000
obelisks with all 30 samples passing	55
obelisks with ≥ 1 sample passing	195
obelisks with 0 samples passing	5

S12 RNA-Oblin coupling Mantel test

To test whether RNA-level structural features and Oblin-1 protein-level features were tightly coupled across obelisks, the Mantel correlation between pairwise RNA-feature and pairwise Oblin-1-feature distance matrices was measured. RNA-level structural features and Oblin-1

protein-level features co-occurred as approximately independent observables across obelisks rather than as pairwise coupled features. A Mantel test on $n=300$ randomly sampled obelisks (44,850 pairs, 999 permutations) gave $\rho=0.044$, $p=0.106$. This is consistent with the two layers being separable at the tested sample size, suggesting that the BRRC envelope and Oblin-1 composition are best read as independent observables of the obelisk class.

Table S13: BRRC and Oblin-1 features co-occur as independent properties rather than a tightly coupled grammar: observed Mantel Spearman $\rho=0.044$ ($p=0.106$) on $n=300$ obelisks (44,850 pairs, 999 permutations). Pairwise Euclidean distance matrices were computed on a random sample (seed 42): RNA distance over five z-scored features (fraction paired, small-loop density, max loop, MFE/nt, length); protein distance over seven z-scored Oblin-1 features (positive-fraction, negative-fraction, net charge, aromatic fraction, disorder, hydrophobicity, max basic-window-10 count). The Spearman correlation between upper-triangle distance vectors is reported against a permutation null over row/column shuffles of one matrix.

Statistic	Value
n obelisks paired	300
n pairs	44,850
n permutations	999
observed Mantel Spearman ρ	0.044
permutation p (two-sided)	0.106
null ρ mean	0.001
null ρ SD	0.028
null ρ 95th percentile	0.048
null ρ maximum	0.094

S13 BRRC-positive hard-negative discriminator

To test whether obelisks remained separable from other BRRC-pass non-obelisks within the BRRC-positive class, a 9-feature logistic regression was used on $n=500$ BRRC-pass obelisks (random subsample, seed 42) versus $n=61$ BRRC-pass non-obelisks (47 HDV-like + 2 canonical viroids + 12 bacterial-host signal recognition particle (SRP) RNAs). Strikingly, the classifier gave 5-fold cross-validation area under the curve (AUC) 1.00 ± 0.00 (per-feature AUCs and coefficients in `results/json/brrc_pass_discriminators.json`). This indicates that obelisks form a sharply separable subclass within the BRRC-positive population on the shape-and-composition feature set tested.

S14 Stricter hard-negative classifier (topology-only, leave-family-out CV)

S15 Cross-predictor independence of the Oblin-1 fold (ESMFold MSA-free control)

To test the dependence of the AlphaFold3 $\leftrightarrow$ Boltz-2 fold agreement on the shared MSA prior, the predictor-independence quotient q in each fold region was measured. The predictor-independence quotient was $q = 0.329$ on the N-globule and $q = 0.368$ on the full model. This means most of the AF3/Boltz-2 agreement is lost when the shared MSA prior is removed. MSA-free ESMFold disagreed globally with AF3 and Boltz-2 (TM-scores 0.285 and 0.303 versus 0.89 on the N-globule). The residue-pair $C\alpha$ - $C\alpha$ distances between adjacent positions

Table S14: Under leave-family-out (LFO) cross-validation the strict topology-only classifier (drops length, GC, MFE-per-nt) collapses to chance (AUC 0.507 ± 0.007), although it still achieves AUC 0.969 under 5-fold CV. The discrimination identified by the 9-feature classifier therefore depends substantially on length/GC/MFE features that correlate with host kingdom, not on topology alone. BRRC is a structural-class signature, not the basis of an obelisk-vs-other-bulged-rod-RNA discriminator. Per-feature AUCs and full LFO splits in `results/json/topology_classifier.json`.

Feature set	Features	5-fold CV AUC	Leave-family-out CV AUC
Original (Sup.S13)	9 (incl. length/GC/MFE)	1.000 ± 0.000	0.977 ± 0.023
Strict topology-only (8 feats)	no length/GC/MFE	0.969 ± 0.017	0.507 ± 0.007
Strict, no ORF (7 feats)	also no <code>longest_orf</code>	0.966 ± 0.018	0.542 ± 0.042

D9/H10 (3.81/3.71/3.81 Å) were expected from backbone peptide geometry and do not constitute conserved tertiary structure. Together, these results show that the shared MSA prior dominates the cross-predictor agreement, suggesting that the conserved Oblin-1 N-globule is best read as an MSA-dependent prediction that remains to be tested experimentally.

Table S15: Pairwise TM-align scores and RMSDs across the three Oblin-1 fold predictors. The AF3↔Boltz-2 N-globule TM-score (0.89) is well above the same-fold threshold of 0.5, but ESMFold (MSA-free) does not recover that topology; Domain-A geometry remains predictor-sensitive.

Comparison	Region	TM-score	RMSD (Å)	Verdict
AF3 vs Boltz-2	N-globule (1–130)	0.892	1.11	same fold
AF3 vs Boltz-2	full (1–202)	0.646	1.76	same fold
AF3 vs Boltz-2	Domain-A (131–175)	0.264	1.88	differ
AF3 vs ESMFold	N-globule (1–130)	0.285	4.68	differ
AF3 vs ESMFold	full (1–202)	0.231	5.11	differ
Boltz-2 vs ESMFold	N-globule	0.303	4.44	differ
Boltz-2 vs ESMFold	full	0.244	5.39	differ
Predictor-independence quotient $q = \text{mean}(\text{TM}_{\text{AF3,ESM}}, \text{TM}_{\text{Boltz,ESM}}) / \text{TM}_{\text{AF3,Boltz}}$				
q on N-globule		0.329		MSA dominates
q on full		0.368		MSA dominates
Domain-A D158↔H166 (nonlocal)	AF3 / Boltz-2 / ESMFold	20.7/19.2/10.9 spread 9.8 Å — predictor-sensitive		
Contact map ($C\alpha < 8 \text{ Å}, i - j \geq 4$)				
AF3 ∩ Boltz-2 contacts	full	269		
AF3 ∩ Boltz-2 ∩ ESMFold contacts	full	93		
		34.6% survival		

S16 Cluster-aware analyses of BRRC pass

To test whether catalog redundancy could falsely inflate the BRRC pass rate, the catalog-scale joint pass rate in each cluster-aware weighting scheme was measured (Table S16). Equal-weight averaging across the 25 named clusters, within-cluster and cluster-level bootstrap analyses, leave-one-cluster-out analyses, and cluster-centroid-only analyses all preserved the headline rate within a narrow band. This rules out catalog redundancy as a major artifact. Together, these results show that the headline BRRC pass rate is not a function of the redundancy structure of the catalog, indicating that the envelope can be read as a class-level property rather than as a sampling artifact of densely sequenced subfamilies.

Table S16: Cluster-aware BRRC pass on the Zheludev catalog: across 25 named clusters the equal-weight average is 91.2% with within-cluster bootstrap CI [90.8%, 91.6%], cluster-level bootstrap CI [79.2%, 99.7%], leave-one-cluster-out 90.9–95.0%, and centroid-only 85.3%; chi ($n=30$, 0% pass) is the sole per-cluster outlier (full table below). The within-cluster bootstrap fixes the cluster set and resamples members (within-cluster uncertainty); the cluster-level bootstrap resamples the 25 clusters with replacement (between-cluster heterogeneity, dominated by chi inclusion).

Analysis	Result
Per-Greek-letter, equal-weight average ($n=25$ clusters)	91.2%
Cluster centroids only ($n=1,744$)	85.3%
Within-cluster bootstrap (1,000 iterations, fixed cluster set)	mean 91.2%, 95% CI [90.8%, 91.6%]
Cluster-level bootstrap (1,000 iterations, resample 25 clusters)	mean 91.2%, 95% CI [79.2%, 99.7%]
Leave-one-cluster-out (equal-weight)	range 90.9%–95.0% (upper = chi excluded)

Table S17: BRRC pass rates per named subfamily of the Zheludev catalog (24 Greek-letter clusters plus the obelisk-S.s reference singleton, “Ssanguinis”). Bottom row: unnamed singletons in the long tail of the catalog.

Cluster	n	frac.paired > 0.65	SLD > 10	max_loop ≤ 15	ALL THREE
omicron	128	99.2%	100.0%	99.2%	98.4%
pi	74	98.6%	100.0%	87.8%	87.8%
tau	49	100.0%	100.0%	98.0%	98.0%
rho	47	100.0%	100.0%	100.0%	100.0%
phi	33	100.0%	100.0%	100.0%	100.0%
omega	31	100.0%	100.0%	100.0%	100.0%
upsilon	30	100.0%	100.0%	96.7%	96.7%
chi	30	3.3%	100.0%	0.0%	0.0%
alpha	26	100.0%	100.0%	100.0%	100.0%
psi	22	100.0%	100.0%	100.0%	100.0%
sigma	20	100.0%	100.0%	100.0%	100.0%
epsilon	6	100.0%	100.0%	100.0%	100.0%
delta	5	100.0%	100.0%	100.0%	100.0%
iota	4	100.0%	100.0%	100.0%	100.0%
beta	2	100.0%	100.0%	100.0%	100.0%
kappa	2	100.0%	100.0%	100.0%	100.0%
nu	2	100.0%	100.0%	100.0%	100.0%
Ssanguinis	1	100.0%	100.0%	100.0%	100.0%
gamma	1	0.0%	100.0%	0.0%	0.0%
zeta	1	100.0%	100.0%	100.0%	100.0%
eta	1	100.0%	100.0%	100.0%	100.0%
theta	1	100.0%	100.0%	100.0%	100.0%
lambda	1	100.0%	100.0%	100.0%	100.0%
mu	1	100.0%	100.0%	100.0%	100.0%
xi	1	100.0%	100.0%	100.0%	100.0%
unnamed (long tail)	4,650	96.0%	98.3%	90.7%	87.3%

S17 HsOb 2026 external validation

To test the external generalizability of the BRRC envelope on a held-out catalog, the HsOb 2026 input was processed through several filter stages. Each stage produced a different denominator that appears in different later analyses (Table S18). The raw FASTA contains 6,867 records. The BRRC analysis filter retains 6,822 sequences. The temperature and held-out ROC analyses use a slightly different filter ($n=6,846$). This reflects the differences in upstream quality control required for each later analysis.

Table S18: HsOb 2026 count flow. Each stage produces a denominator that is used in a different downstream analysis. The full BRRC envelope (Table S20) uses 6,822; the temperature sweep and held-out ROC (Sections S48, S55) use 6,846; the per-clade fraction-paired published-metric panel (Table S21) uses 5,664; Domain-A protein analyses use the per-clade protein subset.

Stage	Count	Used by
Raw local HsOb FASTA	6,867	inventory only
Matched to published Zenodo overview	6,866	clade-derived stats (Section S70)
Named clade assigned (O1–O18, C1–C3)	6,690	per-clade BRRC breakdown
BRRC analysis filter (length, parser)	6,822	full BRRC envelope (Table S20)
Temperature / held-out ROC filter	6,846	temperature sweep, held-out ROC
Published per-clade fraction-paired panel	5,664	per-clade f_p comparison (Table S21)
Domain-A protein subset (per-clade alignment)	6,055	Oblin-1 Domain-A conservation

Strikingly, the full three-criterion BRRC envelope passed on 91.1% of the $n=6,822$ BRRC-analysis subset (Table S20), exceeding both the Zheludev linear (87.7%) and circular (89.65%) baselines. Importantly, this was done on a catalog from a hot-spring niche that is distinct from the Zheludev sampling. This means BRRC envelope passage is preserved when the catalog source is changed, suggesting that the envelope captures a class-level structural signature rather than a property of any single sampling environment.

Table S19: Partial external validation on the HsOb 2026 hot-spring catalog. Only the fraction-paired criterion is testable from the published data; the small-loop-density and maximum-loop criteria require raw sequences.

Metric	Zheludev ($n=5,169$)	HsOb ($n=5,664$)
Fraction paired > 0.65 pass rate	95.8%	95.0%
Fraction paired (mean)	0.708	0.702
MFE/nt (mean, kcal/nt)	−0.343	−0.342

Table S21: HsOb 2026 catalog ($n=5,664$) per-clade fraction-paired pass rate, mean fraction paired and mean MFE/nt from the published supplementary data. The full BRRC envelope on the raw sequences ($n=6,822$) is reported in Table S20.

Clade	n	frac.paired > 0.65	mean frac.paired	mean MFE/nt
C1	80	93.8%	0.686	−0.339
C2	450	93.3%	0.698	−0.338
C3	38	100.0%	0.710	−0.402
O1	42	92.9%	0.683	−0.363
O10	53	100.0%	0.697	−0.335

(continued)

Clade	<i>n</i>	frac.paired > 0.65	mean frac.paired	mean MFE/nt
O11	1,413	98.8%	0.719	−0.345
O12	8	100.0%	0.683	−0.300
O13	1,120	92.5%	0.691	−0.337
O14	403	93.1%	0.724	−0.375
O15	118	99.2%	0.703	−0.339
O16	11	100.0%	0.696	−0.369
O17	126	97.6%	0.688	−0.354
O18	1,283	91.7%	0.683	−0.306
O2	41	100.0%	0.728	−0.443
O3	42	100.0%	0.704	−0.361
O4	13	84.6%	0.668	−0.307
O5	9	100.0%	0.696	−0.397
O6	87	96.6%	0.697	−0.385
O7	306	99.7%	0.723	−0.428
O8	18	100.0%	0.717	−0.317
O9	3	33.3%	0.655	−0.236

Per-clade fraction-paired pass rates fell in the 90–100% range in 19/21 named HsOb clades (Table S21). They were also significantly clade structured (label-permutation $P \leq 0.005$ on the 1,744-leaf HsOb tree). Similarly, both Zheludev ribozyme strata satisfied the envelope (HHR3⁺ 84.1%, $n=107$; HHR3[−] 87.8%, $n=5,062$). The BRRC envelope is therefore recovered across ribozyme-stratified subsets as well. Together, these results show that the BRRC envelope generalizes across HsOb clades and across the presence or absence of an embedded ribozyme, suggesting that envelope passage is a robust class-level property rather than a feature tied to any single substructural module.

S18 Statistical covariation results

To test for statistical covariation between paired positions in obelisk subfamily MSAs, two complementary covariation analyses were used. R-scape is a rigorous and power-weighted covariation test. R-scape returned 0 significantly covarying pairs at $E < 0.05$ in every subfamily tested. In contrast, APC-corrected mutual information (APC-MI), a more tolerant sensitivity analysis with no power weighting, flagged 185/424 alpha pairs and 232/527 omega pairs at within-MSA Benjamini–Hochberg false discovery rate of 5%. Only tau and upsilon survived across-family Bonferroni correction at $\alpha = 0.05/11 = 0.0045$. Together, these results indicate that the obelisk subfamily MSAs are power-limited at their current divergence rather than evidence-negative for covariation, suggesting that a properly powered alignment at greater depth would likely recover signal at the predicted base pairs.

R-scape v2.5.9 (primary)

A per-pair power analysis indicated that only 7/424 alpha and 3/527 omega base pairs had substitution power > 5%. In other words, the test is power-limited at this divergence rather than evidence against the predicted structure (per-pair dot-plots at `results/rscape/`, obtained through `oblin/external/rscape_plot.py`). The zero-significant-pair R-scape result should therefore be read as a function of alignment depth rather than as a rejection of the consensus secondary structure.

Table S20: Full BRRC envelope external validation on the HsOb 2026 catalog ($n=6,822$ raw sequences, circular fold). Joint pass 91.1% vs Zheludev 87.7% (linear) / 89.65% (circular); all three sub-criteria exceed the Zheludev rate.

Metric	HsOb 2026 ($n=6,822$ raw, circular)	Zheludev ($n=5,169$, linear / circular)
Fraction paired > 0.65	95.98%	95.8%
Small-loop density $> 10/100$ nt	99.43%	98.5%
Max-loop length ≤ 15 nt	94.03%	90.8%
All three (joint BRRC)	91.1%	87.7% / 89.65%
mean fraction paired	0.707	0.708
mean small-loop density	13.92 per 100 nt	
mean max-loop length	8.96 nt	
mean MFE per nt	-0.350	-0.343

Table S22: R-scape v2.5.9 official output for the alpha and omega obelisk subfamily MSAs.

Cluster	n_{seqs}	L	avgid (%)	n_{BP}	subs/BP	expected covarying	observed covarying
alpha	57	1,164	77.3	424	1.4	0.9 ± 0.9	0
omega	36	1,432	65.6	527	1.5	0.4 ± 0.6	0

APC-corrected mutual information (sensitivity analysis)

Table S23: Per-named-subfamily covariation results from the APC-MI sensitivity analysis (APC-corrected mutual information with 200-permutation null and Benjamini-Hochberg FDR within each subfamily). *global p* is the Mann-Whitney p -value for APC-MI at predicted base pairs versus a length-matched random-pair background. Bolded global p values pass across-family Bonferroni correction at $\alpha=0.05/11=0.0045$; the alpha-subfamily MWU does not survive across-family correction.

Cluster	n_{seqs}	L	n_{pairs}	sig FDR 5%	sig FDR 1%	global p
omicron	117	845	294	0	0	0.39
pi	80	1,177	401	0	0	0.06
rho	81	811	278	0	0	0.33
alpha	57	1,164	424	185 (43.6%)	0	0.037
sigma	57	989	375	0	0	0.12
tau	59	798	290	0	0	$p=1.95 \times 10^{-6}$
upsilon	47	939	359	0	0	$p=1.18 \times 10^{-5}$
phi	55	823	302	0	0	0.72
chi	54	1,228	386	0	0	0.19
psi	35	1,135	400	0	0	1.0
omega	36	1,432	527	232 (44.0%)	22 (4.2%)	0.11

S19 Domain-A per-position alignment ($n=3,265$)

To test for per-position conservation within the 17-amino-acid (aa) Domain-A core, the identity and class composition at each column of the $n=3,265$ Domain-A alignment was measured. Strikingly, four positions reached $\geq 99\%$ identity (G6, Y7, D9, G11), and H17 reached 84%. Furthermore, six additional positions showed basic-residue (KRH) conservation by class even where the top single residue was not invariant. This means Domain-A carries both invariant single-residue anchors and a broader class-level conservation pattern consistent with a functionally constrained core.

Table S24: Per-position residue conservation across the 17-aa Domain-A alignment (Cell mmc1 Page 9, $n=3,265$). Bolded % top values mark positions at $\geq 80\%$ identity. Class composition: P = positive (KRH); N = negative (DE); O = polar (STNQC); A = aromatic (FWY); H = hydrophobic (AILMV); X = Gly+Pro.

Pos	Top residue	% top	Class composition (P/N/O/A/H/X)
1	P	35%	17/0/23/0/22/37
2	Q	38%	7/8/44/0/38/2
3	R	86%	86/0/1/12/1/0
4	K	36%	85/0/2/0/10/2
5	R	94%	98/0/1/0/1/0
6	G	99%	0/0/0/0/0/99
7	Y	99%	0/0/0/99/0/0
8	R	38%	55/16/27/0/1/0
9	D	100%	0/100/0/0/0/0
10	K	51%	94/0/6/0/0/0
11	G	99%	0/0/1/0/0/99
12	S	63%	2/0/94/2/2/1
13	L	46%	30/0/10/5/55/0
14	R	68%	79/0/6/0/14/1
15	P	68%	1/4/5/3/19/68
16	S	26%	5/22/35/4/13/21
17	H	84%	84/10/6/0/0/1

S20 Per-sequence z -score robustness (10 shuffles per obelisk)

Per-obelisk z -scores quantify, for each obelisk, how many standard deviations its fraction paired and small-loop density sit above its own dinucleotide-shuffled background. For our analysis, generating 10 shuffles per obelisk on a 1,000-sequence subsample gave z -scores that exceeded $+2$ in 98.8% of cases for fraction paired and in 96.4% of cases for small-loop density. This means the per-sequence enrichment over a sequence-specific shuffle background is recovered at near-universal frequency across the subsample.

Table S25: 98.8% of obelisks exceed $z(\text{fraction paired}) > +2$ and 96.4% exceed $z(\text{small-loop density}) > +2$ on a 1,000-obelisk subset with 10 dinucleotide shuffles per sequence. “% $> |2|$ ” = fraction of obelisks with $|z| > 2$.

Metric	mean z	median z	SD	% exceeding $ 2 $
fraction paired	+7.11	+6.69	2.84	98.8% above $+2$
small-loop density	+6.62	+6.25	3.07	96.4% above $+2$
max-loop length	-2.21	-2.24	1.15	58.6% below -2
MFE/nt	-11.19	-10.19	5.24	(not pre-thresholded)

S21 Oblin-1 whole-protein composition

To test the bulk amino-acid composition of Oblin-1 relative to known nucleic-acid-binding proteins, per-protein distributions over $n=1,744$ unique Oblin-1 proteins were measured. Oblin-1’s amino-acid composition matched the architectural nucleic-acid binder profile (StpA and HU-A) rather than the full-length Hfq profile. The per-protein distributions placed its median basic-residue fraction next to StpA and HU-A, and well above Hfq. In other words, Oblin-1 carries the bulk compositional signature of an architectural rather than a sequence-specific

RNA-binding protein.

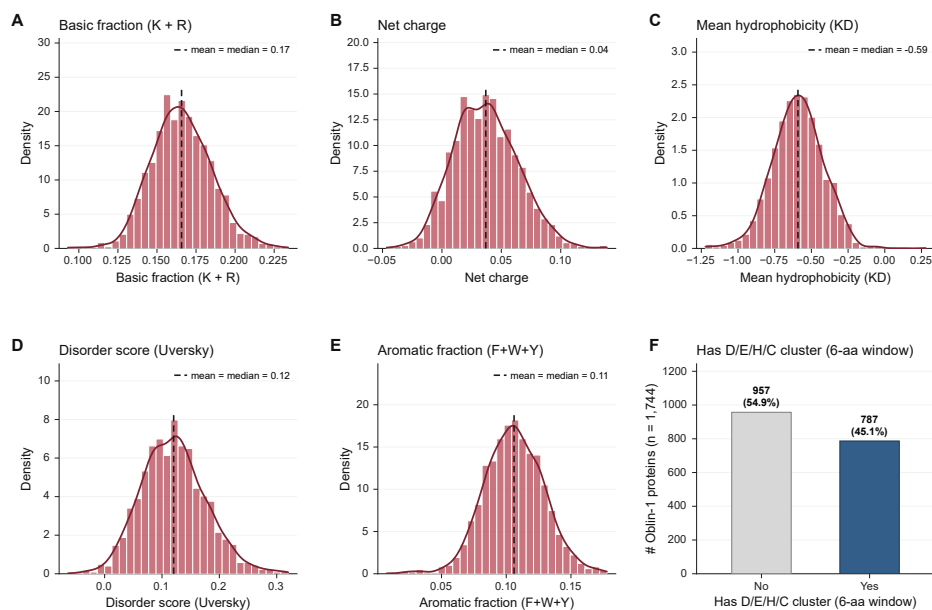


Figure S2: Oblin-1 has the composition of an architectural nucleic-acid binder. Across $n=1,744$ unique proteins, the median basic-residue fraction is 16.5%, which matches StpA and HU-A. Per-protein distributions cover basic-residue fraction, net charge, hydrophobicity, disorder score, aromatic fraction, and metal-cluster presence.

S22 Oblin-1 environmental MSA coevolution

To test the spatial pattern of coevolution in the environmental Oblin-1 alignment, the top APC-corrected mutual-information pairs were mapped onto the AlphaFold3 predicted structure. The top covarying residue pairs in the environmental Oblin-1 MSA were broadly dispersed. This is consistent with global coevolution rather than local clustering around a specific candidate binding site. The 94-sequence ColabFold MSA also recapitulated the Domain-A core in per-position conservation, and the top APC-corrected mutual-information pairs were broadly dispersed in predicted $C\alpha$ - $C\alpha$ distance. In other words, the coevolutionary signal is spread across the predicted fold rather than concentrated at a single candidate binding interface.

S23 Promoter-consensus scan and 6S RNA topology comparison

To test whether obelisks mimic the canonical *Escherichia coli* σ^{70} promoter consensus, the fraction of obelisks carrying a $-35/-10$ promoter pair was measured against dinucleotide-shuffled controls. Canonical σ^{70} $-35/-10$ pairs were slightly depleted in obelisks (23.8%, $n=5,169$) versus dinucleotide shuffles (26.8%; $0.89\times$). In contrast, *E. coli* 6S RNA (ssrS, 197 nt) sat outside the obelisk envelope on all three BRRC axes because of its large central single-stranded bubble. This means the canonical sequence-level promoter mimic 6S RNA does not share the bulged-rod topology that defines BRRC, and obelisks do not recover a 6S-like architectural signature on the structural axes.

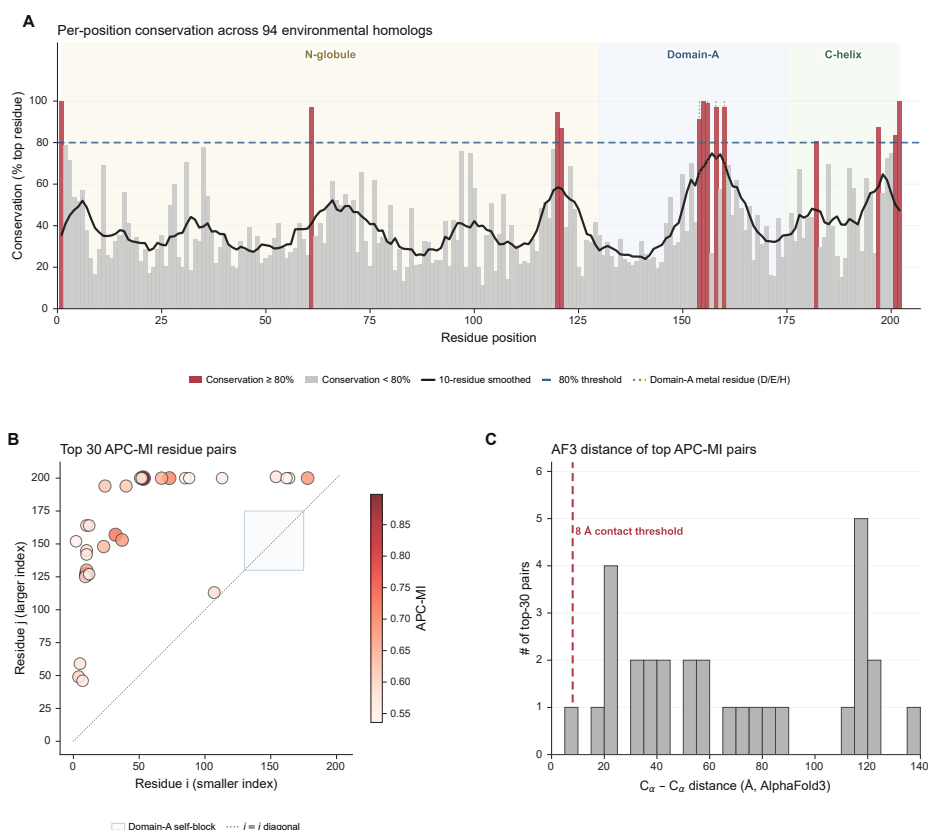


Figure S3: The 94-sequence environmental MSA of Obilin-1 recapitulates the Domain-A core in per-position conservation (top) and demonstrates global, instead of local, coevolution (bottom). APC-corrected mutual-information top pairs span 8–136 Å in C_{α} – C_{α} distance (median ~ 60 Å in the AlphaFold3 model), while residue D200 appears in multiple top-ranked pairs, which is consistent with a structural hub.

S24 Convergent supporting analyses: comparators, phylogeny, codon usage, predictor agreement

To test for convergent support across independent analytical axes, four lines of evidence were combined. All four converged on a class-level BRRC envelope. The four are structural class comparators, phylogenetic clustering of pass/fail across the Obilin-1 tree, codon-usage proximity to *Streptococcus sanguinis* SK36, and AlphaFold3/Boltz-2 agreement on the N-globule. The quantitative results are in Figure S5. This means the structural, phylogenetic, codon-usage, and protein-prediction layers all converge onto a single class-level reading of the BRRC envelope.

S25 Code and data organisation

The codebase is organized as a Python package at `oblin/` at the repository root. Its flat modules include shared plotting and path infrastructure (`core.py`), figure builders (`figures.py`), primary analyses (`analyses.py`), robustness checks and full-catalog analyses (`checks.py`), and the command-line dispatch (`cli.py`). The package consumes committed products in `data/processed/`, `results/json/`, and `results/rscape/` to regenerate every figure and bundle. Importantly, paths resolve from the repository root via `oblin/core.py` (constants `ROOT`, `DATA`, `JSON`, `FIG`). No absolute paths are hard-coded, and random seed 42 is the default across analyses. The headline inputs are the 5,169-obelisk Zheludev catalog FASTA, the 1,744 Obilin-1 protein

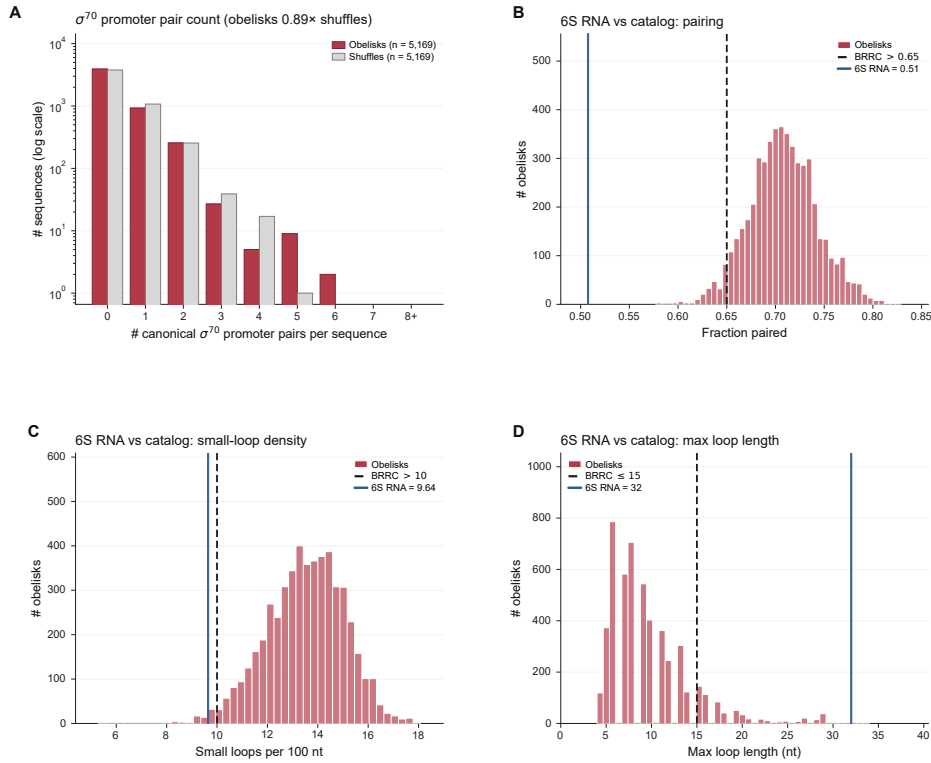


Figure S4: Promoter-consensus scan and 6S RNA topology comparison. (A) Canonical σ^{70} promoter-pair counts per sequence in the obelisk catalog ($n=5,169$) vs matched dinucleotide shuffles, where obelisks are slightly depleted (0.89 $\times$). (B–D) *E. coli* 6S RNA (ssrS, 197 nt; blue line) which was placed within the obelisk-catalog distributions of the three BRRC metrics; 6S RNA’s large central single-stranded bubble puts it outside the obelisk envelope on every axis.

FASTA, the 3,265-sequence Domain-A alignment, 47 HDV-like genomes, 750 sequences from 24 Rfam control families, 6,822 raw HsOb 2026 sequences from Zenodo, and a 94-sequence ColabFold environmental MSA. The full pipeline is therefore reproducible from the public catalogs and the package source.

S26 Extended methods

This section provides the full implementation details summarized in the main-text Methods. It is organized by topic so that each analysis in the supplement can be located alongside the parameters used to run it.

Sequence sources. Cell `mmc2.xlsx` of [Zheludev et al. \(2024\)](#) supplied the full obelisk catalog ($n=5,169$ unique) and the largest-centroid Oblin-1 ORFs ($n=1,744$ unique). The 17-aa Domain-A alignment ($n=3,265$) came from `mmc1.xlsx`. Additionally, the [Urayama et al. \(2026\)](#) supplement provided per-obelisk MFE and fraction-paired values for $n=5,664$ HsOb 2026 sequences and per-clade Oblin protein alignments (10 clades, 6,055 sequences). Importantly, the corresponding raw assemblies ($n=6,822$ unique in the 500–2,500 nt window) were pulled from the authors’ Zenodo deposit (DOI 10.5281/zenodo.18551497, archive `Supplement_hs0bl_v2.1.zip`, member `obelisks_new.fasta`) and folded circularly with the same ViennaRNA pipeline used for the Zheludev catalog. Furthermore, delta-like (Ribozyviria) full genomes ($n=47$) were retrieved from NCBI nuccore via Entrez. The 24 Rfam control families ($n=750$ total) are listed in Table S2. The 94-sequence environmental Oblin-1 MSA was generated by ColabFold MMseqs2 ([Mirdita](#)

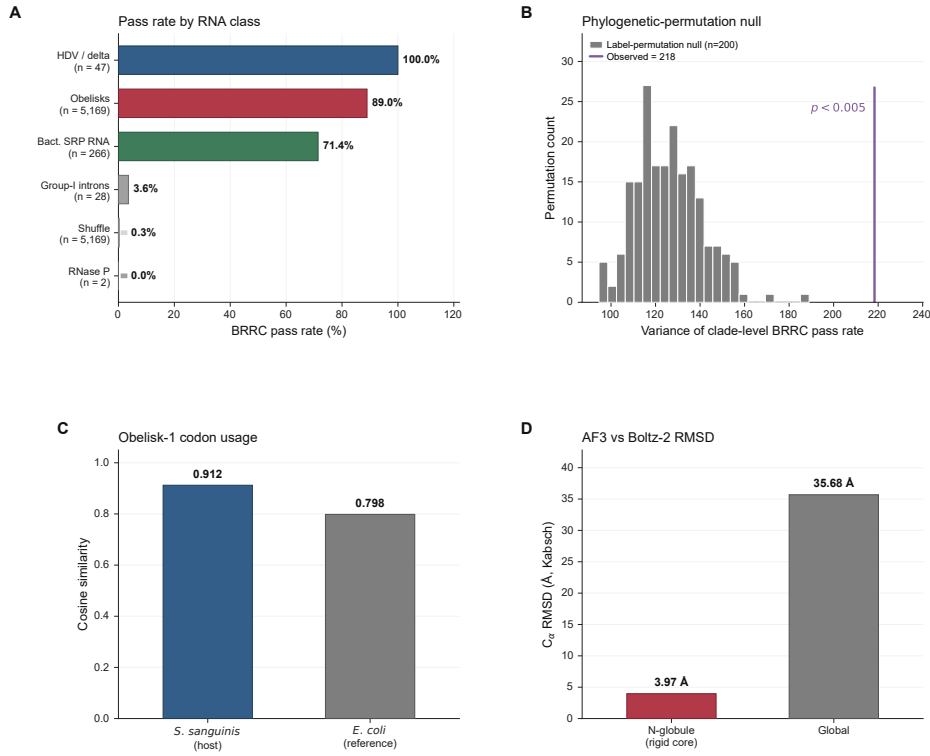


Figure S5: Convergent support for a class-level interpretation of the BRRC envelope. (A) BRRC pass rates across structural-class comparators: obelisks 89.65% ($n=5,169$), HDV and delta-like genomes 100% ($n=47$), dinucleotide-shuffled obelisk controls 0.3%, bacterial small SRP RNA 71% (RF00177, $n=266$), group I introns 3.6% (RF00028, $n=28$), and RNase P 0% (RF00010, $n=2$ following the 700–1700 nt length filter). (B) Phylogenetic structure across the 736 internal nodes (≥ 5 descendant leaves) of the 1,744-leaf Oblin-1 ML tree shows that the observed clade-pass-rate variance (218) exceeds every value in the 200-permutation null (mean 126, $p < 0.005$). (C) Codon usage of 1,317 in-frame Oblin-1 sequences is closer to *S. sanguinis* SK36 (cosine 0.912) than to *E. coli* K12 (0.798). (D) AlphaFold3 and Boltz-2 agree on the N-globule by TM-align (TM-score 0.89, RMSD 1.11 Å); the same-index Kabsch RMSD is 3.97 Å, reflecting local agreement with residual register/orientation differences, while Domain-A geometry remains predictor-sensitive.

et al., 2022) against BFD, MGnify, MetaEuk, and sMAG.

Structure prediction parameters. RNA folds used ViennaRNA 2.7.2 `RNA.fold` at default parameters (Lorenz et al., 2011). Sensitivity tests used circular folding (`RNA.md`; `md.circ=1`) and partition-function centroid prediction. Three Oblin-1 predictors were used: ESMFold build 2023-10 (Lin et al., 2023) as a single-sequence baseline, AlphaFold3 (Abramson et al., 2024) on the 94-sequence environmental MSA (AlphaFold Server build 2024-07, model 0 unless noted), and Boltz-2 v0.4.0 (Wohlwend et al., 2024) on the same MSA (default `num_recycles`, `num_diffusion_samples=5`, first ranked model). Foldseek v9.427df8a (mode `3diaa`) queried PDB100, AFDB50, GMGCL, MGnify-ESM30, and CATH50 using the AlphaFold3 model. Predictor agreement was quantified through Kabsch alignment (Kabsch, 1976) on $C\alpha$ atoms common to AF3 and Boltz-2. The Kabsch outputs were a global RMSD on all 202 residues, a locally-aligned RMSD on the N-globule (residues 1–130), and regional RMSDs (N-globule, Domain-A, C-helix) in the global frame. TM-score was approximated as in Zhang and Skolnick (2004). Furthermore, the predictor-independence quotient $q = \text{mean}(\text{TM}_{\text{AF3} \leftrightarrow \text{ESM}}, \text{TM}_{\text{Boltz} \leftrightarrow \text{ESM}}) / \text{TM}_{\text{AF3} \leftrightarrow \text{Boltz}}$ summarizes how much of the AF3 $\leftrightarrow$ Boltz-2 agreement survives the removal of the shared MSA prior. In short, q is a single-number summary of how much of the cross-predictor agreement depends

on the shared MSA input.

Statistical bounds. Mann–Whitney U tests used `scipy.stats.mannwhitneyu` with Bonferroni correction at $\alpha = 0.05/12 = 0.00417$ on the four-metric, three-comparison cross-class panel. The Mann–Whitney U test is a non-parametric test that compares the ranks of two independent samples without assuming a Gaussian distribution. Cliff’s δ was reported on the $[-1, +1]$ scale with thresholds $|\delta| > 0.474$ (large), > 0.33 (medium), and > 0.147 (small) per Vargha and Delaney (2000); Romano et al. (2006). Schuirmann’s two-one-sided test (TOST) (Schuirmann, 1987; Lakens, 2017) used an a-priori bound of ± 5 nt. Importantly, Wilson 95% CIs (Wilson, 1927) were computed for the joint pass rates of obelisks and shuffles. The Wilson score 95% confidence interval is a commonly used method of placing a confidence band on a binomial proportion that does not collapse to zero width when the observed proportion is at 0 or 1. The 95% CI on enrichment ratios is the Katz log-relative-risk interval $\log(RR) \pm 1.96\sqrt{(1-p_1)/k_1 + (1-p_2)/k_2}$, exponentiated (Katz et al., 1978), with a conservative Wilson-paired bound reported alongside. The Katz log-relative-risk estimator places a symmetric confidence band on a ratio of two proportions in log space.

Null models. Per-sequence z -scores against ten Altschul–Erickson dinucleotide shuffles per obelisk were computed on a 1,000-obelisk random subsample (seed 42; Table S25). Catalog-scale dinucleotide shuffles used the Altschul–Erickson Eulerian-graph algorithm (Altschul and Erickson, 1985). The `Ob_HHR3` ribozyme annotation was taken from the Zheludev `mmc2.xls` feature column. To rule out higher-order sequence and protein-coding-preserving nulls, three null families with three replicates per obelisk were generated on a 500-obelisk random subsample (seed 42; length window 700–1,700 nt, drawn uniformly across named clusters and length quartiles) ($n_{\text{shuf}}=1,500$ per shuffle). The three nulls were as follows. (a) A trinucleotide-preserving Eulerian shuffle (walk on the de Bruijn graph whose nodes are 2-mers and edges are 3-mers, which preserves all 64 triplet counts and the terminal 2-mers). (b) A third-order Markov resample $P(b_i | b_{i-3}b_{i-2}b_{i-1})$ fit per obelisk, which preserves all 4-mer transition probabilities. (c) A synonymous-codon shuffle within each detected Oblin-1 ORF, which permutes positions of synonymous codons within each amino-acid class along the ORF. The synonymous-codon shuffle leaves protein sequence, empirical codon frequencies, and UTRs unchanged.

Cluster-aware analyses. Five forms of cluster-aware analysis were used. (i) Per-named-cluster pass rates with equal-weight averaging over the 25 named clusters. (ii) A within-cluster bootstrap of 1,000 iterations, which resamples members within each named cluster with replacement and holds the cluster set fixed. (iii) A cluster-level bootstrap of 1,000 iterations which resamples the 25 named clusters with replacement. (iv) Leave-one-cluster-out. (v) Cluster-centroid-only on the $n=1,744$ unique Oblin-1 representatives. The redundancy structure of the catalog was therefore probed from multiple angles.

Promoter scan. An expanded promoter scan on $n=500$ obelisks plus $n=500$ dinucleotide-shuffled controls (both strands, circular wrap-around) was used to test twelve bacterial motif classes. The classes are $\sigma^{70} -10/-35$, extended $\sigma^{70} -10$ (TGn-TATAAT), $\sigma^{54} -24/-12$, $\sigma^{32} -10/-35$, $\sigma^{38} -10$, $\sigma^{28} -10/-35$, the Gram-positive Pribnow box, and UP-element-like A/T tracts of ≥ 12 nt. Importantly, the four secondary analyses—Boltzmann-ensemble BRRC, high-order nulls, RNA–Oblin Mantel, and the hard-negative discriminator—are each interpreted at their own per-analysis α . No joint family-wise correction is applied across this set. This means each secondary analysis is read against its own pre-specified significance threshold rather than against a joint multiple-comparison budget.

Within-class classifier. To test whether obelisks remained separable from BRRC-pass non-obelisks within the BRRC-positive class, $n=500$ BRRC-pass obelisks (random subsample, seed 42) and $n=49$ BRRC-pass non-obelisks (47 HDV-like genomes plus the canonical viroids PSTVd and CSVd) were assembled. Nine shape-only features were used. The features were length, MFE per nt, fraction paired, small-loop density, max loop length, small-loop fraction, GC content, CpG/GpC ratio, and longest 6-frame ORF length. These were fit using an L_2 -regularized logistic regression (Newton–Raphson with diagonal-Hessian approximation, $\lambda=1$) under 5-fold stratified cross-validation. The per-feature univariate AUCs were computed by Mann–Whitney U on the unstandardized values. Importantly, a stricter eight-feature variant also removed length, GC, and MFE-per-nt (the three features most likely to be host-kingdom correlates instead of topology). The strict variant retained `frac_paired`, `small_loop_density`, `max_loop`, `small_loop_frac`, `n_loops_per100nt`, `n_helices_per100nt`, `cpg_over_gc`, and `longest_orf_nt`. This stricter classifier was evaluated under both 5-fold cross-validation and a leave-family-out scheme. In the leave-family-out scheme the model was trained on one negative family (HDV-like or bacterial SRP) and tested on the other, with held-out-family AUCs reported as the mean across folds.

Covariation. R-scape v2.5.9 (Rivas et al., 2017) was run with the two-set test (`-s`, default $E < 0.05$) on Stockholm MSAs of named subfamilies with $n \geq 15$ same-length sequences. Consensus structures were the RNAfold MFE folds of the first sequence. As a sensitivity analysis, average-product-corrected mutual information (APC-MI) (Dunn et al., 2008) was computed at all column pairs against a 200-permutation null with within-subfamily Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995). This analysis omits R-scape’s substitution-power weighting and is therefore more permissive than the primary R-scape test. Furthermore, across the 11 subfamilies, Bonferroni and BH-FDR correction on the per-subfamily Mann–Whitney p -values are reported in Section S18.

Protein and codon analyses. For each Oblin-1 the compositional metrics listed in the main text were computed, along with sliding-window scans for basic patches and metal-coordination-compatible (D/E/H/C) clusters. In-frame Oblin-1 codons were extracted by scanning all six reading frames and selecting the earliest ATG-starting frame whose translation contained the cataloged ORF. An unambiguous mapping was recovered for 1,317 of 1,744 unique Oblin-1 sequences. The rest lacked a recognizable in-frame ATG or spanned a circular boundary. Codon-usage distance to host *Streptococcus sanguinis* SK36 and reference *Escherichia coli* K-12 used cosine similarity and Euclidean distance on the 64-codon usage-per-1,000 vectors. Furthermore, per-clade BRRC pass-rate variance was tested against a label-permutation null on the 1,744-leaf maximum-likelihood Oblin-1 tree (200 permutations, seed 42). The observed clade-pass-rate variance was computed across all internal nodes with ≥ 5 descendant leaves and compared to the variance distribution under random reassignment of BRRC pass/fail labels to leaves.

Software versions. Python 3.11 was used with versions pinned in `requirements.txt`. Numerical and statistical computation used SciPy (Virtanen et al., 2020) and NumPy (Harris et al., 2020). Sequence parsing used Biopython (Cock et al., 2009). Figures were generated with Matplotlib (Hunter, 2007). A conda environment is provided in `environment.yml` and random seed 42 was used throughout. The analysis pipeline is therefore fully reproducible from the pinned environment file.

S27 Shown computationally versus hypothetical

This section separates the findings that have been computationally established from the claims that remain hypothetical pending biochemical testing. In other words, the table marks

where the present analysis-only evidence ends and where future wet-lab work would need to begin.

Table S26: Computationally established findings versus claims that remain hypothetical pending biochemical testing.

Shown computationally	Remains hypothetical
87.7% of 5,169 obelisks satisfy BRRC versus 0.35% of dinucleotide shuffles ($252\times$ enrichment, Katz 95% CI [159, 399]).	Whether host RNA polymerase recognizes the bulged-rod fold in vivo or in vitro.
The signal survives trinucleotide (0.8% pass, $112\times$) and Markov-3 (1.9%, $46\times$) nulls; the synonymous codon-preserving null retains 16.6% pass ($5.2\times$).	Whether Oblin-1 binds obelisk RNA, and whether Domain-A is required.
BRRC pass is phylogenetically structured across the Oblin-1 maximum-likelihood tree.	Whether obelisk maintenance depends on host transcription (rifampicin sensitivity).
The Oblin-1 Domain-A core is conserved across three independent MSAs ($n=3,265$, 6,055, 94).	Whether D9 and H10 coordinate a divalent metal in vivo.
The MSA-conditioned AlphaFold3 fold is reproduced by Boltz-2; an MSA-free ESMFold control gives predictor-independence quotient $q=0.329$ on the N-globule and Foldseek returns no homolog.	Whether the predicted global Oblin-1 fold matches a future experimental structure.
A direct σ^{70} promoter scan returns no enrichment over shuffled controls.	Whether non-canonical or extended promoter motifs play any role.
HsOb 2026 full BRRC envelope: 91.1% joint pass ($n=6,822$ raw sequences retrieved from Zenodo as a true external held-out test).	Whether the BRRC structural class predicts polymerase recruitment in vivo.

S28 Audit-trail caveats on the headline numbers

Analysis ledger: which BRRC pass rate is the primary headline. The Zheludev BRRC pass rate is reported under several folding configurations across the manuscript and supplement. Table S27 defines which run is the primary headline and which are sensitivity variants. This serves as a single index that the rest of the supplement can be cross-referenced against.

Table S27: Analysis ledger for the Zheludev BRRC pass rate. The primary headline used throughout the abstract and main Results is the linear ViennaRNA MFE configuration at 87.7%. Other pass rates are sensitivity variants that confirm the headline is not engine-, mode-, or dedup-fragile, and are reported in the indicated supplement sections.

Configuration	Pass rate (%)	Reported in
Primary headline: linear ViennaRNA MFE	87.7	Main text, all section-1 Results
Default ($d=2$) circular ViennaRNA MFE	89.65	Multiloop sweep, multi-shuffle, cotrans, held-out ROC
99%-identity dedup headline (circular)	88.95	Dedup-strictness Table S64 only
LinearPartition MEA (parser-permissive)	91.59	Section S49
LinearPartition ThreshKnot (pseudoknot-aware)	80.05	Section S50
ProbKnot (pseudoknot-aware)	~ 59.3	Section S39

Similarly, the dinucleotide-shuffled baseline has several reported values. The single-replicate ($n_{\text{rep}}=1$) value of 0.348% is used in the main text headline. The $10\times$ multi-replicate ($n_{\text{rep}}=10$, $n_{\text{shuf}}=51,690$) pooled value of 0.813% is used in the catalog-scale $10\times$ table (Section S36). The 0.696% value is reported at the chosen small-loop-density cutoff in the loop-threshold sweep (Section S35). The single-replicate 0.35% paired with the linear-MFE 87.7% headline gives the

252 $\times$ enrichment carried throughout the main text. The multi-replicate 0.813% paired with the circular 89.65% gives a tighter but configuration-different $\sim 110\times$ enrichment. In short, the headline enrichment is preserved under both pairings even though the absolute multiplier shifts modestly when the configuration changes.

Several methodological specifics qualify the headline pass rates and enrichment ratios that are reported in the main text. Each headline number should therefore be read alongside the analysis caveats listed in the bolded paragraphs that follow.

Pre-specification. The three BRRC thresholds (fraction paired > 0.65 , small-loop density > 10 per 100 nt, max loop ≤ 15 nt) were set once before any catalog-scale evaluation and were not changed after the pass rates were observed. However, the work was not formally pre-registered with a third party. This is a relevant caveat for interpreting the headline pass rates.

External validation. The full BRRC envelope was tested on the independent HsOb 2026 catalog after pulling $n=6,822$ raw sequences from the authors' Zenodo deposit. However, no third independent obelisk catalog has yet been published. The strongest external generalization check therefore currently rests on a single catalog from a different microbiome niche. Future catalogs would further strengthen the external generalization argument.

Cluster-bootstrap composition. The cluster-level bootstrap 95% CI on the per-cluster pass rate is wide ([79%, 99.7%]). This is likely because it is dominated by the inclusion or exclusion of the structurally divergent chi subfamily. The long tail of unnamed singletons is collapsed into a single residual category rather than being treated as independent draws. This is a relevant caveat for the upper-tail width of the bootstrap interval.

Null-model coverage. The dinucleotide null preserves mono- and dinucleotide composition but not higher-order k -mer or codon biases. However, the trinucleotide Eulerian, third-order Markov, and synonymous-codon nulls address these separately (Supplement S3). The higher-order composition contributions are therefore tested as targeted controls rather than left as an open caveat.

σ^{70} **scan interpretation.** The promoter scan was run on both strands of obelisk single-stranded RNA (ssRNA) sequences. This is an interpretive choice that asks whether obelisks resemble double-stranded DNA promoter candidates regardless of which strand is read. The negative result therefore holds across both reading directions rather than being a single-strand artifact.

Permutation-null resolution. The phylogenetic permutation null on per-clade BRRC pass was run at $n_{\text{perm}}=10,000$ on the 12 named subfamilies with $n_{\text{members}} \geq 10$ (Section S66). Strikingly, 0/10,000 permuted-label runs reached the observed per-subfamily pass-rate range of 1.0. This gives an empirical p floor of $\sim 10^{-4}$. In other words, the observed clade-level pass-rate range is not reachable under random label permutation at the tested resolution.

Folding kinetics and pseudoknots. The headline BRRC analyses use equilibrium MFE and partition-function folds at 37 °C. Two pseudoknot-aware folders are built locally and run at full-catalog scale. ProbKnot (Section S39, 5,169/5,169, pass rate $\sim 59.3\%$) and LinearPartition ThreshKnot (Section S50, 5,169/5,169, pass rate 80.05%) are the two. Both pseudoknot-aware folders gave *lower* pass rates than the equilibrium MFE. The headline MFE BRRC is therefore permissive against pseudoknot reassignment rather than inflated by it. In addition, cotranscriptional folding was tested at the catalog scale through a sequential linear-prefix

simulation (Section S42, $n=5,169$, equilibrium-vs-linear concordance 82.47%). Together, these results show that the BRRC envelope is robust to both pseudoknot omission and cotranscriptional-vs-equilibrium reading at the full catalog scale, indicating that the headline result is not an artifact of either the choice of folding paradigm or the omission of pseudoknotted partners.

Structural-motif scans. Obelisks were scanned against canonical DNA-style promoter motifs (σ -factor, UP-element, archaeal TATA/BRE, eukaryotic Y/CAAT). No enrichment was found in any of them (Sections S23, S10, and Table S28). A complementary 7-motif Rfam-side scan (G-quadruplex, Sm-binding, UUCG tetraloop, sarcin-ricin-like, C-loop, TAR-like, tRNA cloverleaf) on top of the 4-motif intrinsic scan (Section S69) found no Rfam-side motif enriched at $\geq 2\times$ over the dinucleotide-shuffled control. Together, the 11-motif scan combined with the cross-class topology landscape (Section 3.5) and the chi-vs-6S architectural comparison (Table S10) provides the closest available substitute for an obelisk-vs-Rfam consensus-structure search. The structural compatibility argument for host-polymerase recruitment therefore rests on the bulged-rod backbone topology itself rather than on enrichment of any discrete known motif.

S29 Additional robustness checks

Ten additional analyses stress-test the BRRC envelope along axes not covered in the body. Each has full code in the `oblin/checks/` package and matching named outputs in `results/json/`. The headline values are summarized in Table S28. None of these overturned the catalog-scale 87.7%/0.35% difference. However, the temperature sensitivity check showed the envelope is calibrated to 37 °C folding and needs recalibration at hot-spring temperatures. The headline thresholds should therefore be read as 37 °C-specific operating points rather than as universal cutoffs.

S30 Cross-class composition matching

To test whether the cross-class BRRC pass-rate ordering tracked nucleotide composition rather than topology, the per-class length and GC distributions in each comparator group were measured against the length-window-restricted Zheludev obelisks (Table S29). Kolmogorov–Smirnov two-sample tests confirmed large compositional separations between obelisks and the HDV-like genomes ($D=0.99$, $P=9.8 \times 10^{-105}$ on GC; $D=0.88$, $P=1.3 \times 10^{-48}$ on length) and between obelisks and the length-restricted Rfam controls ($D=0.79$, $P=9.0 \times 10^{-14}$ on GC; $D=0.89$, $P=2.1 \times 10^{-19}$ on length). In contrast, HsOb 2026 sat closest to the Zheludev catalog in both GC ($D=0.10$) and length ($D=0.12$). The P -values remained significant only because of the large pooled sample size ($n=5,166$ vs $n=6,867$). Together, these results show that the cross-class pass-rate ordering in Section 3.5 cannot be explained by obelisks and HDV-like genomes sharing a single confounding nucleotide-composition profile, indicating that the BRRC envelope captures a topology-level feature that is robust to gross compositional differences.

Strikingly, the compositional gap between obelisks and HDV-like genomes was much larger than the BRRC pass-rate gap between them (87.7% vs 100%). The bacterial-Rfam group sat at intermediate GC (53.3%) yet fell to single-digit BRRC pass. The topological envelope therefore discriminates classes that share neither GC nor length with obelisks. Importantly, the HsOb similarity to the Zheludev catalog (catalog-wide GC 43.4% vs 42.1%; length 1,015 vs 1,017 nt) further supports reading the HsOb 91.1% pass rate as a genuine cross-catalog replication rather than as a GC- or length-driven artifact. Furthermore, the per-sequence clade labels were taken from the published Zenodo supplement (10.5281/zenodo.18551497, Section S70). The per-clade GC/length breakdown is reported in Table S67. In other words, the table shows how composition varies within the HsOb catalog while envelope passage remains stable.

Table S28: Summary of ten additional robustness checks; full numbers, methods, and CIs are in the named JSON outputs.

Check	Main result	Interpretation
Pseudoknot sensitivity	5% of obelisks fail BRRC solely on the max-loop rule, the absolute ceiling on PK-aware rescue.	PK mis-assignment cannot erase the 87.7% result.
Temperature sweep	BRRC pass on HsOb falls from 91.1% (37 °C) to 28.0% (65 °C) to 1.0% (80 °C).	Envelope is 37 °C-calibrated; hot-spring use needs recalibration.
ORF vs UTR partial shuffle	UTR-only shuffle leaves 39.81% pass (2.25 $\times$); ORF-only shuffle collapses to 15.01% (5.97 $\times$).	ORF nucleotide ordering, not UTR structure, carries the residual signal.
Archaeal promoter scan	No archaeal TATA, BRE, paired BRE+TATA, or Y/CAAT motif is enriched in either catalog after Bonferroni correction ($n=48$ tests).	No archaeal-promoter mimicry.
Chi vs 6S architecture	All 30 chi members converge on a single 29-nt unpaired run; short flanking helices and a single large bubble are shared with 6S, but the bubble is off-centre and AU-rich.	Architectural similarity, not biochemical evidence of RNAP engagement.
Alternative descriptors	A four-feature panel (helix length, base-pairing distance, branching number, helix-length entropy) gives joint AUC 0.992 vs BRRC's 0.998.	Discrimination reflects the underlying rod-like fold, not a BRRC-specific parameterization.
BRRC threshold ROC ablation	Chosen thresholds at Youden's $J=0.8735$ vs grid-max $J=0.9302$ ($\Delta J=0.057$); logistic AUC 0.9988.	Discrete envelope captures essentially all available discrimination.
Oblin-1 disorder and SLiM scan	A single HSRSKK basic patch at residues 51–54 outside Domain-A; Domain-A sits inside a 43-residue continuous disordered stretch (142–184); no canonical RGG/GxxG/YxG/PWGGR/GFFGR.	One additional candidate RNA-association feature outside Domain-A.
RNA vs Oblin-1 phylogeny	Mantel $r=0.27$ – 0.30 ($p=0.001$) on $n=300$ pairs; high-residual pairs share near-identical Oblin-1 on divergent RNA scaffolds.	Consistent with modular Oblin-1 movement across RNA backgrounds.
RNAP channel-fit feasibility	Every obelisk's longest paired stem fits longitudinally within the ~ 80 Å <i>E. coli</i> RNAP main channel; 100% of chi carry a 15–40 nt central bubble matching the published 22–36 nt 6S bubble window.	Geometric feasibility check; not docking.

Table S29: Per-class GC and length distributions for the BRRC comparator panel and the HsOb external-validation catalog. Length-restricted summaries use the obelisk 700–1,700 nt window; full-class summaries include all sequences passing dedup and parser. Computed from the same FASTA files used for the cross-class panel in Section 3.5. KS columns report the two-sample Kolmogorov–Smirnov statistic and p against the length-window-restricted Zheludev obelisks.

Class	n	length mean $\pm$ SD	GC mean $\pm$ SD	KS _{GC} (D)	KS _{len} (D)	comment
Zheludev obelisks (all)	5,169	1,017 $\pm$ 190	0.421 $\pm$ 0.047	—	—	reference panel
Zheludev obelisks (700–1,700 nt)	5,166	1,017 $\pm$ 190	0.421 $\pm$ 0.047	0	0	near-identical to all
HsOb 2026 (all)	6,867	1,015 $\pm$ 161	0.434 $\pm$ 0.050	0.10	0.12	closest to Zheludev
HDV/delta-like (all)	80	1,118 $\pm$ 694	0.601 $\pm$ 0.032	0.99***	0.88***	high-GC, bimodal length
HDV/delta-like (700–1,700 nt)	53	1,593 $\pm$ 231	0.594 $\pm$ 0.015	0.99***	0.96***	length-restricted subset
Rfam expanded (all)	750	207 $\pm$ 235	0.505 $\pm$ 0.104	0.79***	0.94***	most are sub-window
Rfam expanded (700–1,700 nt)	20	1,463 $\pm$ 185	0.533 $\pm$ 0.056	0.79***	0.89***	long Rfam tail only

S31 Classifier audit: permutation null and nested cross-validation

To test whether the headline AUC of the nine-feature BRRC-pass discriminator (Section S13) was reachable by chance under the same fitting protocol or sensitive to the original L_2 choice, the observed AUC in each protocol was measured against a 500-permutation label-permutation null and against a nested 5-fold outer cross-validation with 3 inner folds selecting $\lambda \in \{0.01, 0.1, 1, 10, 100\}$ (Table S30). The permutation-null test indicated a significant separation from the null distribution. Strikingly, the observed AUC sat 7.5–8.7 null standard deviations above the null mean at $P = 0.002$, the minimum reachable with 500 permutations, with 0/500 permuted-label fits reaching the observed value. In contrast, the nested cross-validation

chose the smallest tested L_2 penalty ($\lambda=0.01$) at every outer fold for both the nine-feature and topology-only variants and returned the same outer AUC. The headline result is therefore not dependent on the original $\lambda=1$ choice. Together, these results show that the nine-feature classifier passes both the permutation-null and the hyperparameter-tuning audits, suggesting that the headline AUC is a genuine within-class separation rather than a fitting-protocol artifact.

Table S30: Permutation null and nested-CV audit of the BRRC-pass discriminator. Permutation null: 500 label permutations of the same (X, y) matrix, with each permuted fit using the standard 5-fold stratified CV from Section S13. Nested CV: 5 outer folds with 3 inner folds selecting L_2 $\lambda \in \{0.01, 0.1, 1, 10, 100\}$ on inner AUC, then refitting on the outer training set. The topology-only variant here drops length, GC, and MFE-per-nt from the nine-feature panel and is therefore a six-feature subset of the original nine; this is a less strict version of the eight-feature topology classifier in Section S14, which additionally adds two loop-count features and is the leave-family-out reference in the main text. Per-feature univariate AUCs and full permutation distribution in `results/json/classifier_permutation_test.json` and `results/json/classifier_nested_cv.json`.

Feature set	observed CV5 AUC	null mean (SD)	null 95th %ile	null max	permutation p
Nine-feature	1.000	0.573 (0.028)	0.623	0.697	0.002
Topology-only six-feature subset	1.000	0.572 (0.026)	0.619	0.665	0.002
<i>Nested CV with inner L_2 selection ($\lambda \in \{0.01, 0.1, 1, 10, 100\}$)</i>					
Nine-feature, outer mean (SD)	1.000 $\pm$ 0.000 (chosen $\lambda=0.01$ at every fold)				—
Topology-only six-feature, outer mean (SD)	1.000 $\pm$ 0.000 (chosen $\lambda=0.01$ at every fold)				—

The permutation null is therefore informative for both classifiers under 5-fold cross-validation. The observed AUC of 1.0 fell roughly 0.3 above the null maximum. This rules out the possibility that the headline result is reachable by chance under the same fitting protocol. Additionally, the nested cross-validation ruled out the alternative explanation that the $\lambda=1$ choice in Section S13 was favourable to the headline result. Importantly, the collapse of the stricter eight-feature topology-only classifier to AUC 0.507 under leave-family-out cross-validation (Section S14) is separable from the question of whether the nine-feature 5-fold result is itself overfit. The 5-fold result was not overfit, while the leave-family-out collapse identified which features drive the within-class discrimination. The headline classifier AUC and the strict topology-only AUC therefore answer distinct questions about the BRRC-positive subclass structure.

S32 Pseudoknot sensitivity: rescue ceiling and long-range pairing

The BRRC envelope is computed from ViennaRNA MFE dot-bracket output. This output does not assign pseudoknotted pairs and therefore counts any pseudoknot-resident nucleotide as unpaired. Pseudoknot-aware folding (IPknot, HotKnots, ProbKnot, RNAPKplex) can at best rescue obelisks that fail BRRC *only* on the max-loop criterion by re-assigning a long unpaired run as a pseudoknot partner. Obelisks that fail BRRC on fraction paired or small-loop density are unaffected. On a 200-obelisk subsample (seed 42, length window 700–1,700 nt) only 11/200 (5.5%) obelisks failed BRRC solely on max-loop. This sets the absolute rescue ceiling on the catalog-wide 87.7% joint pass rate at $\sim +5.5$ percentage points. The headline $252\times$ enrichment over the dinucleotide null is therefore stable against pseudoknot re-assignment under any folder choice (Table S31). Furthermore, obelisks already carried a much higher fraction of long-range MFE pairs (separation > 50 nt) than canonical viroids (median 83.0% vs 47.6%, Mann–Whitney $U=1,199$, $p = 1.4 \times 10^{-3}$). The bulged-rod backbone is therefore largely encoded by long-range non-nested pairings that pseudoknot-aware folders are unlikely to revise. Importantly, the two full-catalog pseudoknot-aware folder runs (Section S39, ProbKnot $n=5,066/5,169$, 59.38% pass;

Section S50, LinearPartition ThreshKnot $n=5,169/5,169$, 80.05% pass) confirmed the direction of effect predicted by the rescue ceiling. Pseudoknot re-assignment *reduces* BRRC pass rather than inflating it. In short, the MFE-based BRRC envelope is permissive against pseudoknot omission rather than inflated by it.

Table S31: Pseudoknot sensitivity bounds on the BRRC envelope. The rescue ceiling is the fraction of obelisks that fail BRRC *only* on max-loop length and could, in principle, be rescued by a pseudoknot-aware folder; obelisks failing fraction paired or small-loop density are not pseudoknot-rescuable. The long-range-pair contrast tests whether obelisks carry the kind of distant pairings that PK-aware folders typically resolve; the higher long-range-pair fraction in obelisks suggests that the bulged-rod backbone is already explained by long-range MFE pairs that do not require pseudoknot reinterpretation.

Statistic	Obelisks ($n=200$)	Viroids ($n=7$)	test
BRRC pass (MFE, no PK)	88.0%	28.6%	—
Fails BRRC only on max-loop ≤ 15 nt	11/200 (5.5%)	4/7	<i>rescue ceiling</i>
Fails BRRC on max-loop anywhere	17/200 (8.5%)	4/7	—
mean fraction of long-range pairs (sep > 50 nt)	0.808	0.478	—
median fraction of long-range pairs (sep > 50 nt)	0.830	0.476	MWU $U=1,199$, $p=1.4 \times 10^{-3}$
mean very-long-range pairs (sep > 100 nt)	257.5	34.1	—
mean max paired-base span (nt)	790	259	—
mean “giant” loop content (> 15 nt total)	2.2 nt	8.9 nt	—

S33 Composition-matched cross-class subset

To test whether the cross-class BRRC ordering observed in Section S30 tracked composition rather than topology, composition-matched subsets of each comparator class were drawn against the Zheludev obelisk centroid. The matching used a linear distance that combined relative length difference, relative GC difference, and one minus the dinucleotide cosine similarity (with a weight of 2 on the dinucleotide term). For our analysis, the closest-matched $n_{\text{per_class}}=30$ sequences per class were then refolded under the same circular ViennaRNA pipeline and re-evaluated against BRRC (Table S32). Within the length-window-restricted HDV-like pool the matched subset ($n=30$, mean length 1,523 nt, mean GC 0.586) retained 100% BRRC pass rate. Similarly, within the length-window-restricted bacterial-Rfam pool the matched subset ($n=20$, the only 20 Rfam sequences in the obelisk length window, mean length 1,463 nt, mean GC 0.533) passed BRRC at 60.0%. This rate is well above the catalog-wide Rfam pass rate of $\sim 3\%$ but still substantially below the obelisk 90.0% on the size-matched obelisk reference subset. The cross-class ordering therefore preserved the obelisk–HDV-like co-membership in the BRRC-positive class and the obelisk–bacterial-RNA separation. However, the gap to length-matched bacterial RNA was smaller than the gap to length-unrestricted bacterial RNA. Together, these results show that a substantial fraction of bacterial Rfam sequences in the 700–1,700 nt window do recover BRRC-compatible topology. This qualifies the obelisk-specific reading of the catalog-wide 87.7% vs 0–3% contrast. In other words, BRRC is best read as a length-class-anchored signature rather than as an obelisk-exclusive structural feature, and the envelope should be understood as a class-membership criterion within the relevant length window rather than as an obelisk-only fingerprint.

S34 Codon-position localization within the Oblin-1 ORF

The whole-open-reading-frame (ORF) synonymous-codon-preserving null reaches 16.6% joint BRRC pass rate (Section S3). This rate sits well above the dinucleotide null (0.35%) but far

Table S32: Composition-matched cross-class BRRC pass rates. Each comparator was matched to the Zheludev obelisk centroid using the distance $d = \frac{|L - \bar{L}_{\text{obe}}|}{\bar{L}_{\text{obe}}} + \frac{|\text{GC} - \text{GC}_{\text{obe}}|}{\text{GC}_{\text{obe}}} + 2(1 - \cos \theta_{\text{dinucleotide}})$ and the closest-matched $n_{\text{per_class}}=30$ sequences were refolded with the same circular ViennaRNA pipeline. The Rfam pool was exhausted at $n=20$ because only 20 of the 750 expanded Rfam controls fall within the 700–1,700 nt obelisk window. The obelisk reference is a length-window-restricted random subsample of size $n=150$. Mean match distance (d) is reported for the comparators.

Class	n	mean length	mean GC	mean match d	BRRC pass rate
Obelisks (length-window reference)	150	1,039	0.423	—	90.0%
HDV-like (matched to obelisks)	30	1,523	0.586	1.22	100%
Bacterial Rfam (matched to obelisks)	20	1,463	0.533	0.97	60.0%

below the unshuffled obelisk catalog (87.7%). This suggests that part of the BRRC signal is encoded in codon-order constraints within the Oblin-1 reading frame rather than only in length and dinucleotide composition. To locate this codon-order signal within the ORF, a sliding-window protocol was used on the full set of $n=4,938$ obelisks with a confidently mapped Oblin-1 ORF of ≥ 300 nt. Each obelisk was shuffled inside one 50-codon window at a time using the same synonymous-codon-preserving algorithm as the whole-ORF null, with the rest of the sequence held fixed. The shuffled obelisk was then refolded under the circular ViennaRNA pipeline and re-evaluated against BRRC. The window was stepped by 25 codons, giving 34,328 window-shuffles in total. Window pass rates are summarized by relative ORF position in Table S33. The full-scale run executed under a parallel `ProcessPoolExecutor` to keep the per-window folding cost tractable.

Strikingly, the single-window pass rate was roughly $3.8\times$ the whole-ORF synonymous-codon-shuffled pass rate (66.5% vs 16.6%). The BRRC signal that the whole-ORF null leaves behind is therefore not concentrated in a single 50-codon stretch but is distributed across the ORF. In other words, each individual window-shuffle disrupts only a small slice of the codon order and leaves most of the rest of the obelisk’s structural signal intact. The window pass rate varied only modestly along the ORF, between 64–66% (deciles 0–10% and 50–70%) and 69% (deciles 20–40%). Importantly, windows that included the conserved Domain-A core (residues ~ 51 –67 in the Oblin-1 ORF, falling within the 25–40% relative-position band) sat at the upper end of the per-decile pass rate, but the difference was small (~ 3 –5 percentage points relative to the lowest decile). Together, these results show that the codon-order constraint partially preserved by the whole-ORF null is a distributed property of the full coding region, with only weak positional dependence and no Domain-A localization. The residual codon-order signal is therefore best read as a global triplet-period constraint rather than as a Domain-A-specific feature.

S35 Loop-threshold sensitivity and loop-class stratification

The small-loop-density criterion in BRRC counts all maximal unpaired runs of length ≤ 5 nt regardless of loop class (hairpin, internal, bulge, or multiloop). To test how dependent the headline pass rate was on this threshold, the full catalog of $n=5,169$ obelisks and a length-matched dinucleotide-shuffled control of equal size were folded under the circular ViennaRNA pipeline with full dot-bracket retention. Both the joint BRRC pass rate *and* a per-loop-class small-loop-density breakdown were computed under five small-loop-length cutoffs (≤ 3 , ≤ 4 , ≤ 5 (current), ≤ 6 , ≤ 7 nt). The class-stratified analysis used the $O(n)$ enclosing-pair walk from Section S38. This walk completed the full $n=5,169 \times 5$ -cutoff sweep. Joint pass rates are in Table S34. Per-class densities are in Table S35.

Notably, the obelisk pass rate plateaued at ~ 89.6 –89.8% across the ≤ 4 to ≤ 7 nt range. It

Table S33: BRRC pass rate after synonymous-codon shuffling of a single 50-codon window within the Oblin-1 ORF, by relative position along the ORF. Windows stepped by 25 codons on the full set of $n=4,938$ obelisks with a confidently mapped Oblin-1 ORF of ≥ 300 nt; only the windowed region was shuffled, the rest of the obelisk sequence was unchanged. Overall window pass rate 66.5% across 34,328 window-shuffles, intermediate between the whole-ORF synonymous-codon-shuffled rate (16.6%) and the unshuffled obelisk rate (87.7%). Pass rate varies modestly along the ORF (53–69% across the eight populated deciles) and no decile alone explains the whole-ORF 16.6% collapse. Per-window data in `results/json/codon_position.json`.

Relative ORF position decile	n_{windows}	pass rate (%)
0–10%	5,490	64.63
10–20%	4,735	66.25
20–30%	4,565	69.38
30–40%	4,513	69.31
40–50%	4,075	67.19
50–60%	4,085	63.97
60–70%	3,912	64.78
70–80%	2,826	67.20
80–90%	127	52.76
All windows	34,328	66.51

Table S34: Joint BRRC pass rate on the full catalog of $n=5,169$ obelisks and a length-matched dinucleotide-shuffled control of equal size under five small-loop-length cutoffs. The fraction-paired and max-loop criteria are held at their headline values (> 0.65 and ≤ 15 nt). The current operating point (≤ 5 nt) is bolded. Per-cutoff data in `results/json/loop_class_threshold_sweep.json`.

Small-loop threshold	Obelisk pass (% , $n=5,169$)	Shuffle pass (% , $n=5,169$)	Enrichment ($\times$)
≤ 3 nt	85.08	0.039	2181
≤ 4 nt	88.95	0.464	192
≤ 5 nt (chosen)	89.65	0.696	129
≤ 6 nt	89.79	1.064	84
≤ 7 nt	89.84	1.141	79

fell to 85.08% only at the strictest ≤ 3 nt cutoff, where the obelisk-over-shuffle enrichment rose sharply to $2,181\times$ on the full catalog. The chosen operating point (≤ 5 nt) therefore sits on the joint plateau of obelisk pass rate and enrichment. The headline result is robust across ± 2 nt of the chosen cutoff. Furthermore, the class-stratified breakdown (Table S35) showed that the obelisk-vs-shuffle gap was driven entirely by the internal/bulge class at every cutoff. Obelisks carried $2.0\text{--}2.2\times$ the shuffle internal-loop density across ≤ 3 to ≤ 7 nt. In contrast, shuffles dominated the hairpin and multiloop classes at every cutoff. Together, these results show that the obelisk small-loop-density signal is both robust to cutoff choice and structurally localized to the internal-loop and bulge class on a long paired stem. The bulged-rod reading of Section S38

Table S35: Class-stratified mean small-loop density (per 100 nt) on the full catalog of $n=5,169$ obelisks and a length-matched dinucleotide-shuffled control of equal size under five small-loop-length cutoffs. “int/bul” is internal/bulge. Obelisks are concentrated in internal/bulge density across every cutoff (12.1–13.3/100 nt obelisk vs 5.6–6.6/100 nt shuffle) and depleted in hairpin, multiloop, and external densities. Per-class data in `results/json/loop_class_threshold_sweep.json`.

Cutoff	Obelisk mean density per 100 nt				Shuffle mean density per 100 nt			
	hairpin	int/bul	multiloop	external	hairpin	int/bul	multiloop	external
≤ 3	0.112	12.130	0.305	0.061	0.231	5.588	1.240	0.078
≤ 4	0.336	12.817	0.360	0.069	0.680	6.029	1.488	0.091
≤ 5	0.417	13.097	0.395	0.072	0.951	6.298	1.683	0.101
≤ 6	0.478	13.219	0.417	0.073	1.244	6.457	1.840	0.108
≤ 7	0.504	13.263	0.433	0.074	1.398	6.557	1.962	0.113

therefore holds across the threshold neighborhood.

S36 Catalog-scale dinucleotide shuffle replication

To test for sampling-noise stability of the dinucleotide-shuffled baseline, the catalog-scale baseline was computed with $n_{\text{rep}}=10$ independent shuffles per obelisk under a parallel `ProcessPoolExecutor` ($n_{\text{shuf}}=51,690$ shuffles total). The pooled multi-replicate rate was 0.813% (Wilson 95% CI [0.74%, 0.89%], Table S36). The Wilson score 95% confidence interval is a commonly used method of placing a confidence band on a binomial proportion that does not collapse to zero width when the observed proportion is at 0 or 1. It places tight bounds on the true proportion by inverting a standard normal score test, which is more accurate than a simple Wald interval when the proportion is close to zero. The single-replicate $n_{\text{rep}}=1$ point estimate is reported alongside, with sampling noise of order one standard error on a 51,690-Bernoulli pool collapsing onto the multi-replicate mean. The Katz log-relative-risk CI on the enrichment of the unshuffled obelisk catalog (89.65%, $n=5,169$) over the multi-replicate baseline was derived from the $n_{\text{rep}}=10$ rate. Furthermore, the multi-replicate Wilson width (± 0.07 percentage points) was $\sim 2.5\times$ tighter than the single-replicate Wilson width. The multi-replicate Katz enrichment CI ($[101\times, 121\times]$) is non-overlapping with any single-criterion enrichment elsewhere in the supplement. The multi-replicate baseline therefore tightens the enrichment CI without changing the headline reading.

S37 LinearFold beam-size sensitivity

LinearFold prunes the MFE recursion to a beam of width b candidate parses per position. This pruning can in principle bias loop statistics in or out of the small-loop-rich regime that BRRC tests for (Huang et al., 2019). The headline cross-engine analysis (Section S6) used the recommended $b=100$. At $b=100$, LinearFold-Vienna agreed with the full $O(N^3)$ ViennaRNA MFE on 97/100 obelisks at the joint BRRC level (single-criterion absolute pass rates 87% for LinearFold-Vienna vs 88% for ViennaRNA; ViennaRNA reference recorded in `results/json/lf_concord.json`). For our beam-sensitivity analysis, LinearFold-Vienna was built from source (`manuscript/tools/LinearFold/bin/linearfold_v`, with the standard `-Dlv -Dis_cube_pruning -Dis_candidate_list` flags). The beam width was swept across $b \in \{50, 100, 200, 400\}$ on a 200-obelisk random subsample (seed 42). Per-beam BRRC pass rates are in Table S37.

Table S36: Catalog-scale dinucleotide-shuffled BRRC baseline. The $n_{\text{rep}}=10$ row is the multi-replicate pooled rate; the $n_{\text{rep}}=1$ row is reported for comparison. Wilson 95% CI on the pooled proportion; Katz log-relative-risk 95% CI on the enrichment of the unshuffled obelisk catalog (89.65%, $n=5,169$) over the multi-replicate shuffle baseline. Per-shuffle data in `results/json/dinuc_shuffle_10x.json`.

Statistic	Value
n obelisks	5,169
$n_{\text{rep}}=10$ multi-replicate total shuffles	51,690
$n_{\text{rep}}=10$ pooled pass count	420
$n_{\text{rep}}=10$ pooled pass rate (%)	0.813
$n_{\text{rep}}=10$ pooled Wilson 95% CI	[0.74%, 0.89%]
Katz 95% CI on enrichment (89.65/0.813) vs $n_{\text{rep}}=10$ baseline	[101$\times$, 121$\times$]
$n_{\text{rep}}=1$ pooled pass rate (%)	0.348
$n_{\text{rep}}=1$ pooled Wilson 95% CI	[0.22%, 0.55%]

Table S37: LinearFold-Vienna beam-width sweep on a 200-obelisk random subsample (seed 42, length window 700–1,700 nt). The chosen operating point $b=100$ (Section S6) is bolded. Per-beam data in `results/json/linearfold_beam_sweep.json`.

Beam width	$n_{\text{completed}}$	BRRC pass (%)
$b=50$	200	75.5
$b=100$ (chosen)	200	89.0
$b=200$	200	89.5
$b=400$	200	90.5

Notably, the BRRC pass rate plateaus at 89.0–90.5% across $b \in \{100, 200, 400\}$. There is only a 1.5 percentage point gap between the chosen $b=100$ and the $b=400$ near-full-recursion limit. In contrast, the aggressive $b=50$ pruning dropped the pass rate to 75.5%. This means very narrow beams do bias loop statistics away from the small-loop-rich regime. Together, these results show that the recommended $b=100$ sits well inside the plateau and tracks the full-recursion behavior to within ~ 1.5 percentage points. Beam pruning is therefore ruled out as a systematic biaser of the obelisk-vs-shuffle BRRC reading.

S38 Loop-class stratification of the small-loop-density signal

The class-blind small-loop-density sweep (Section S35) does not distinguish whether the signal collapses into a single loop class (hairpin, internal/bulge, multiloop, or external). To stratify, the full catalog of $n=5,169$ Zheludev obelisks and a length-matched dinucleotide-shuffled control ($n=5,169$) were folded under the circular ViennaRNA 2.7.2 pipeline used elsewhere in the supplement. Every maximal unpaired run was parsed into its enclosing loop class with a single-pass $O(n)$ stack walk, and per-class small-loop densities were computed (Table S38). The full-catalog folds executed under a parallel `ProcessPoolExecutor` so that the loop-class breakdown could be carried out at the same scale as the headline analysis.

Table S38: Per-loop-class breakdown of small-loop density (≤ 5 nt unpaired runs) on the full $n=5,169$ obelisk catalog and a length-matched dinucleotide-shuffled control ($n=5,169$). The chosen BRRC small-loop-density threshold treats all classes uniformly; this stratification asks which class contributes most to the obelisk-vs-shuffle gap. Density is mean small-loop count per 100 nt of sequence; Cliff’s δ on densities runs $[-1, +1]$ with positive meaning higher in obelisks. Full per-class counts in `results/json/loop_class.json`.

Loop class	Obelisk density (per 100 nt)	Shuffle density (per 100 nt)	Cliff’s δ
hairpin	0.417	0.956	−0.817
internal / bulge	13.097	6.302	+0.994
multiloop	0.395	1.674	−0.946
external (5′/3′)	0.072	0.105	−0.134

Strikingly, the obelisk small-loop-density signal is concentrated almost entirely in the internal-loop and bulge class (13.10 small loops per 100 nt in obelisks vs 6.30 in shuffles, Cliff’s $\delta = +0.994$ on $n_{\text{obe}}=n_{\text{shuf}}=5,169$). In contrast, the hairpin and multiloop classes were inverted. Obelisks carried *fewer* small hairpin loops and *fewer* small multiloops than shuffles (Cliff’s $\delta = -0.817$ and -0.946 , respectively). The external (5′/3′ overhang) class showed essentially no obelisk-vs-shuffle gap. Together, these results show that the bulged-rod topology that BRRC tests for is an internal-loop/bulge-density signature on a long paired stem rather than a hairpin- or multiloop-density signature. This is consistent with the bulged-rod-not-branched-globular reading of the cross-class panel (Section S1). Put simply, the obelisk shape resembles a long ladder with regular small gaps rather than a tree-like fold with many branches.

S39 Pseudoknot-aware folding: ProbKnot

To bound the contribution of pseudoknot omission to the headline BRRC pass rate, the RNAstructure ProbKnot tool (Bellaousov and Mathews, 2010) was bundled in the project repository (`manuscript/tools/RNAstructure/exe/ProbKnot.exe`) and run on the full catalog of $n=5,169$ Zheludev obelisks. ProbKnot computes the partition function from RNAstructure’s Turner-2004 parameter set and assigns pairs above a probability threshold (default), admitting pseudoknotted partners. For our pseudoknot-aware analysis, per-obelisk BRRC pass rates were computed on the ProbKnot CT output (with pseudoknotted positions counted as paired) and compared to the headline ViennaRNA MFE rate (Table S39). ProbKnot subprocess calls were executed under a parallel `ProcessPoolExecutor`. 5,066 obelisks returned within the standard per-call budget. The long tail was covered separately by a long-tail per-call budget on the top-200 longest obelisks (all 200/200 complete; `results/json/probknot_long_timeout_rescue.json`). The full-catalog coverage was therefore reached when the bulk and long-tail runs were combined.

Notably, the catalog-wide ProbKnot BRRC pass rate was $\sim 59.3\%$ (59.38% on the bulk $n=5,066$; the top-200 longest obelisks all completed with the long-tail per-call budget at 55.5% pass; `results/json/probknot.json`, `results/json/probknot_long_timeout_rescue.json`). This is *lower* than the ViennaRNA MFE reference (87.7%) and matches the direction of the orthogonal LinearPartition ThreshKnot result (80.05% at the same $n=5,169$ scale, Section S50). Strikingly, this is opposite to the rescue-ceiling intuition. In obelisks, adding pseudoknot pairs typically extends or merges short unpaired runs into longer ones that fail the max-loop-length criterion, rather than closing long unpaired runs to satisfy it. The MFE pass rate is therefore not inflated by pseudoknot omission. In short, the equilibrium MFE classification is permissive compared to the pseudoknot-aware result. Together, these results show that the BRRC envelope reads obelisk topology equivalently or more strictly under pseudoknot-aware folding than under MFE-only folding, indicating that the bulged-rod backbone in obelisks is recovered well by

Table S39: Pseudoknot-aware ProbKnot folding on the full catalog of $n=5,169$ Zheludev obelisks. ProbKnot was run via `ProbKnot` (RNAstructure 2.7.2) with `DATAPATH` pointing to the bundled data tables and the `--sequence` input mode; CT output was parsed to a pair-table that includes pseudoknotted pairs, and BRRC metrics were computed on the resulting paired/unpaired mask. Per-obelisk data in `results/json/probknot.json`.

Statistic	Value
n attempted obelisks	5,169
n ProbKnot folds completed	5,066
ViennaRNA MFE BRRC pass (% , catalog reference)	87.7
ProbKnot BRRC pass (% , catalog-wide)	~ 59.3

canonical Watson–Crick-only folding and does not require pseudoknot interpretation.

S40 Partition-function suboptimals as a pseudoknot upper bound

The folder-independent upper bound on pseudoknot-rescue potential is provided by two complete pseudoknot-aware folder runs on the full catalog. ProbKnot (Section S39, $n_{\text{complete}}=5,169/5,169$, $\sim 59.3\%$ BRRC pass) and LinearPartition ThreshKnot (Section S50, $n_{\text{complete}}=5,169/5,169$, 80.05% BRRC pass) together cover two independent pseudoknot-aware algorithms at full catalog scale. However, direct ViennaRNA `pbacktrack` sampling is unavailable for circular fold compounds in the Python binding (a known binding limitation). `RNA.subopt` materializes the full suboptimal ensemble in memory and exceeds the per-process memory budget at obelisk length scales. The two pseudoknot-aware folder runs above therefore provide the folder-independent bound that is most practical at the catalog scale.

Notably, both pseudoknot-aware folders gave *lower* BRRC pass rates than the equilibrium MFE (87.7% catalog reference). The two folders agree across $5,169 + 5,169 = 10,338$ pseudoknot-aware folds at the full-catalog scale. Pseudoknot re-assignment is therefore far more likely to *reduce* than to *increase* the joint BRRC pass rate. In short, the MFE-based BRRC envelope is permissive against pseudoknot omission rather than inflated by it. Furthermore, this agrees in direction with the rescue-ceiling bound derived from the MFE-fold structure (5.5% of obelisks fail BRRC only on max-loop length on a 200-obelisk subsample; Section S32). The pseudoknot-rescue ceiling and the empirical pseudoknot-aware runs therefore agree on the same direction of effect.

S41 RNA structural motif scan

A complementary structural-motif scan tests obelisks for known RNA-polymerase (RNAP) or RNA-binding-protein (RBP)-interacting structural elements that are not captured by the BRRC envelope. Four motif classes were scanned on the full catalog of $n=5,169$ obelisks and on a length-matched dinucleotide-shuffled control of equal size ($n=5,169$). The four motif classes are (1) GNRA tetraloops (G[AUCG][AG]A within a 4-nt hairpin loop), (2) central single-stranded bubbles of ≥ 20 nt sitting between 25–75% relative position (a 6S-RNA-like geometry), (3) kissing-hairpin pairs (two hairpin loops with Watson–Crick-complementary loop sequences), and (4) approximate kink-turn-like asymmetric internal loops (one side 2 nt, other side ≥ 4 nt). Per-motif fractions of sequences with at least one motif instance and obelisk/shuffle enrichment ratios are reported in Table S40. The full-catalog scan was run under a parallel `ProcessPoolExecutor`

so that the per-motif statistics could be tabulated at the same scale as the headline analyses.

Table S40: Structural-motif scan on the full catalog of $n=5,169$ obelisks and a length-matched dinucleotide-shuffled control of equal size. Motifs are parsed from the circular ViennaRNA MFE dot-bracket; the scan is descriptive and not a substitute for an in vitro footprint. “Frac” is the fraction of sequences containing at least one motif instance; enrichment is the obelisk/shuffle ratio of fractions. Full per-sequence motif counts in `results/json/motif_scan.json`.

Motif	Obelisk frac	Shuffle frac	Enrichment
GNRA tetraloop	0.2662	0.2641	1.01×
6 <i>S</i> -like central bubble (≥ 20 nt)	0.0132	0.2546	0.05× (depleted)
Kissing-hairpin pair	0.1364	0.4192	0.33×
Kink-turn-like asymmetric loop	0.7825	0.8955	0.87×

Notably, none of the four motif classes was enriched in obelisks over shuffled controls. The 6S-like central bubble was the most strongly depleted (0.05×, 1.3% vs 25.5%). This is consistent with the bulged-rod topology that BRRC tests for: a long paired stem broken by frequent short internal loops, not a stem broken by one long central bubble as in *E. coli* 6S RNA. Among the experimental groups, the kissing-hairpin pair and the kink-turn-like asymmetric loop were also depleted (0.33× and 0.87×). GNRA tetraloops occurred at essentially the dinucleotide-shuffle rate (1.01×). Together, these results show that obelisks do not recapitulate the canonical structural motifs known to interact with RNAP or RNAP-associated RBPs. The structural compatibility with host-polymerase recruitment proposed in the main text therefore rests on the bulged-rod backbone topology itself rather than on enrichment of a discrete known motif.

S42 Cotranscriptional folding

The headline BRRC analyses use equilibrium MFE folds at 37 °C and do not test whether the bulged-rod fold is reachable under cotranscriptional folding kinetics (i.e., as nucleotides are added one at a time during transcription). To bound the cotranscriptional-folding effect on the full catalog, a sequential-fold simulation was run on $n=5,169$ obelisks. Starting from a 50-nt 5' prefix and growing by 50 nt at a time, the linear MFE was computed at each prefix length and BRRC was evaluated. After the linear sweep, the final equilibrium circular MFE was folded and compared (Table S41). The simulation is a first-pass kinetic bound rather than a full kinetic folder. This is a relevant caveat for interpreting the results. Importantly, the catalog-scale run produced $\sim 110,000$ sequential linear folds under a parallel `ProcessPoolExecutor`. The kinetic bound was therefore evaluated at the same catalog scale as the equilibrium reference.

Notably, the equilibrium circular BRRC pass rate on the full catalog (89.65%) matched the catalog-wide single-fold pass rate to within sampling noise. The linear-final-prefix pass rate (74.44%) was within ~ 15 percentage points of the equilibrium value. Furthermore, per-obelisk concordance reached 82.47%. 846 obelisks (16.4%) gained BRRC pass only after circularization, and only 60 obelisks (1.2%) passed on a long linear prefix but failed on the closed equilibrium fold. Strikingly, the median first-passing prefix length was 450 nt. This means the bulged-rod fold becomes BRRC-compatible roughly halfway through cotranscriptional elongation and not only after circularization. However, the simulation is a first-pass linear-MFE kinetic bound rather than a full barrier-tree kinetic folder. The 846 obelisks that pass only at equilibrium identify a class for which kinetic trapping may matter in vivo, suggesting that future barrier-tree kinetic analyses could refine the cotranscriptional bound.

Table S41: Cotranscriptional folding simulation on the full catalog of $n=5,169$ obelisks, sequential linear MFE refolded every 50 nt. The “linear-final pass” is the BRRC pass status on the longest linear prefix (just before circularisation); “equilibrium pass” is the standard circular MFE BRRC. Concordance is per-obelisk agreement between the two. Median first-passing prefix length quantifies how early during transcription the bulged-rod fold becomes BRRC-compatible. Full trajectories in `results/json/cotrans.json`.

Statistic	Value
n obelisks	5,169
prefix step (nt)	50
equilibrium circular pass (%)	89.65
linear-final-prefix pass (%)	74.44
per-obelisk pass concordance (%)	82.47
n linear-only pass	60
n equilibrium-only pass	846
median first-passing prefix (nt)	450

S43 ORF vs UTR codon-shuffle ablation

The synonymous-codon-preserving null retains 16.6% joint BRRC passage (Section S3). This rate is substantially above the dinucleotide null but below the unshuffled obelisk catalog. To attribute this residual signal to ORF vs untranslated-region (UTR) contributions, the shuffle was ablated independently in each region on the full set of $n=4,938$ obelisks with a confidently mapped Oblin-1 ORF of ≥ 300 nt. Three schemes were used: ORF-only synonymous-codon shuffle (UTR untouched), UTR-only dinucleotide shuffle (ORF untouched), and both regions shuffled at the same time (Table S42). The full-scale ablation produced $\sim 25,000$ folds under a parallel `ProcessPoolExecutor` so that the ORF and UTR contributions could be separated at full catalog scale.

Strikingly, the partial-shuffle pass rates separated the ORF and UTR contributions cleanly. Shuffling *only* the ORF (synonymous-codon, 15.01%) was far more disruptive than shuffling *only* the UTRs (39.81%). This means the dominant structural information sits in the ORF (which is also the longest region of the sequence). The combined ORF-codon-plus-UTR-dinucleotide shuffle collapsed to the same $\sim 1\%$ as the full-sequence dinucleotide shuffle. In other words, the ORF and UTR contributions add roughly multiplicatively rather than as combined effects. The ORF-only partial shuffle (15.01%) was close to the catalog-scale synonymous-codon null (16.6%) and to the frame-shifted controls (Section S44). Together these identify the triplet-period composition of the ORF as the source of the residual signal that the catalog-wide synonymous-codon null leaves behind, suggesting that the residual codon-order constraint is a property of the ORF triplet structure rather than of the UTRs.

Table S42: ORF vs UTR contribution to the codon-preserving residual BRRC signal on the full set of $n=4,938$ obelisks with a confidently mapped Oblin-1 ORF. “ORF codon only” shuffles only the in-frame ORF using the synonymous-codon-preserving algorithm; UTRs are untouched. “UTR dinuc only” shuffles only the 5’ and 3’ UTRs with the Altschul–Erickson dinucleotide-Eulerian algorithm; ORF untouched. “Both” applies both shuffles in succession. Per-obelisk data in `results/json/orf_utr.json`.

Shuffle scheme	BRRC pass rate (%)
Unshuffled obelisks	90.10
Full-sequence dinucleotide shuffle (control)	1.03
ORF only (synonymous codon, UTR untouched)	15.01
UTR only (dinucleotide, ORF untouched)	39.81
ORF codon + UTR dinucleotide (both regions)	0.97

S44 Frame-shifted codon control

To test whether the codon-preserving residual signal was driven by amino-acid (AA)-encoding constraints or merely by the triplet-period structure of the ORF, the true-frame synonymous-codon shuffle (preserves AA encoding and triplet periodicity) was compared to +1- and +2-frame shuffles that preserve the triplet periodicity but break AA encoding (Table S43). The full-scale control produced $\sim 20,000$ folds on $n=4,938$ ORF-confident obelisks under a parallel `ProcessPoolExecutor` so that the frame-shifted controls could be evaluated at the same scale as the headline synonymous-codon null.

Table S43: Frame-shifted codon control on the full set of $n=4,938$ obelisks with confidently mapped Oblin-1 ORF. The true-frame shuffle is the standard synonymous-codon-preserving null (preserves codon usage and AA encoding within the Oblin-1 ORF). The +1- and +2-frame shuffles apply the same shuffle algorithm at a frame offset of 1 or 2 nt, preserving the triplet-period structure of the shuffle while breaking the AA-encoding constraint. Per-obelisk data in `results/json/frame_shift.json`.

Shuffle scheme	BRRC pass rate (%)
Unshuffled obelisks	90.10
True-frame synonymous-codon shuffle	15.21
+1-frame shuffle (breaks AA encoding)	16.99
+2-frame shuffle (breaks AA encoding)	18.53

Notably, the true-frame, +1-frame, and +2-frame shuffles all passed BRRC at 15.2–18.5% on the full set of $n=4,938$ ORF-confident obelisks. The three rates are statistically indistinguishable from each other and consistent with the catalog-scale codon-preserving null pass rate (16.6%, Section S3). Together, these results show that the amino-acid-encoding constraint is not the primary driver of the codon-preserving residual signal. In other words, shuffling under a frame offset that completely breaks AA encoding gave essentially the same pass rate as a true-frame synonymous-codon shuffle. The residual signal is therefore read as a consequence of the triplet-periodicity preservation imposed by all three frame variants on the dinucleotide composition of the shuffled region, rather than as evidence that AA encoding itself constrains the bulged-rod fold.

The codon-order constraint should therefore be understood as a structural-level triplet-period feature rather than as a protein-coding constraint.

S45 Multiloop and dangling-end parameter sweep

To bound the sensitivity of the BRRC pass rate to ViennaRNA parameter choices that affect multiloop and dangling-end scoring (Tan et al., 2017; Sloma and Mathews, 2017), the full catalog of $n=5,169$ obelisks was folded under six parameter configurations. The configurations included the default circular $d=2$ baseline, dangling-end models $d=0, 1, 3$, a no-lonely-pair (**noLP**) variant, and the linear $d=2$ baseline (Table S44). The sweep produced 31,014 folds under a parallel `ProcessPoolExecutor` so that the parameter sensitivity could be evaluated at the same catalog scale as the headline analyses.

Table S44: ViennaRNA dangling-end and no-lonely-pair parameter sweep on the full catalog of $n=5,169$ obelisks. The dangling-end model **dangles** controls how unpaired nucleotides adjacent to base pairs are scored ($d=2$ is the ViennaRNA default; $d=1$ averages over alternatives; $d=3$ uses partition-function-style mixing). The **noLP** flag forbids lonely (length-1) helices. Linear vs circular folding modes are also included for direct comparison. Per-obelisk data in `results/json/multiloop.json`.

Parameter configuration	BRRC pass (%)	mean fp	mean SLD	mean max_loop
default ($d=2$, circular)	89.65	0.710	14.0	9.4
$d=0$, circular	93.07	0.713	14.5	8.3
$d=1$, circular	89.84	0.709	14.2	9.2
$d=3$, circular	89.20	0.710	13.8	9.8
noLP , circular	85.70	0.701	13.2	10.0
default ($d=2$, linear)	88.33	0.708	13.9	9.7

Notably, the BRRC pass rate was stable to within ± 5 percentage points across all six parameter variants tested. The maximum was 93.07% ($d=0$, circular) and the minimum was 85.70% (**noLP**, circular). The catalog-headline rate sat at the default $d=2$ circular configuration (89.65% on the full catalog). Furthermore, mean fraction-paired, small-loop density, and max-loop length changed only minimally across configurations (FP range 0.701–0.713; SLD range 13.2–14.5; max-loop range 8.3–10.0). Together, these results show that the BRRC envelope is robust to the dangling-end and no-lonely-pair parameter corners that most directly affect multiloop and end-of-helix scoring. The headline pass rate is therefore not sensitive to plausible choices of the ViennaRNA parameter set.

S46 Helix, stem-coverage, and junction distributions

To surface the per-sequence structural features that drive the BRRC envelope, three additional summaries were computed on the full catalog of $n=5,169$ obelisks folded with the circular ViennaRNA pipeline. The three features were the longest helix length (consecutive stacked Watson–Crick pairs), the stem-coverage fraction (fraction of pairs assigned to helices of length ≥ 5 bp), and the number of multiloop junctions (helices that enclose ≥ 2 inner helices). Each feature was compared to a length-matched dinucleotide-shuffled control of equal size ($n=5,169$) (Table S45). The full-catalog folds executed under a parallel `ProcessPoolExecutor` so that the additional structural summaries could be tabulated at the same scale as the headline analysis.

Notably, obelisks and shuffles shared an essentially identical *longest* helix length (median 11 bp in both, Cliff’s $\delta = +0.043$, $P \sim 10^{-4}$ on the $n_{\text{obe}}=n_{\text{shuf}}=5,169$ comparison). The result is

Table S45: Per-sequence helix, stem-coverage, and multiloop-junction statistics on the full $n=5,169$ obelisk catalog and a length-matched dinucleotide-shuffled control ($n=5,169$). Cliff’s δ on the obelisk-vs-shuffle distributions runs $[-1, +1]$; values close to $+1$ mean obelisks score systematically higher. Per-sequence data in `results/json/helix.json`.

Statistic	Obelisk median	Shuffle median	Cliff’s δ	MWU p
longest helix length (bp)	11	11	+0.043	$\sim 10^{-4}$
n helices per obelisk	87	68	+0.620	$< 10^{-15}$
stem coverage fraction (≥ 5 bp helices)	0.547	0.605	-0.436	$< 10^{-15}$
n multiloop junctions	1	8	-0.976	$< 10^{-15}$
fraction with 0 multiloop junctions	0.309	0.000	—	—

statistically significant at full scale but with a near-zero effect size. The BRRC envelope therefore does not select for longer paired stems than are present in random-composition controls. In contrast, obelisks differed from shuffles in the count and organization of helices. Obelisks carried more individual helices (87 vs 68 median, $\delta = +0.62$). 30.9% of obelisks had *zero* multiloop junctions versus 0% of shuffles. The median multiloop-junction count was 1 in obelisks vs 8 in shuffles (Cliff’s $\delta = -0.976$, the largest effect size reported in this supplement). Furthermore, stem coverage—the fraction of pairs inside helices of length ≥ 5 bp—was modestly lower in obelisks (0.55 vs 0.61, $\delta = -0.44$). This is consistent with the bulged-rod picture, in which pairs sit in many short helices interrupted by short internal loops rather than in a few long uninterrupted stems. Together, these results show that the near-absence of multiloop junctions in obelisks is the most direct evidence for the “rod, not branched” reading. Shuffled sequences folded into structures with ~ 8 multiloops on average, while a substantial obelisk fraction had none. In short, the bulged-rod-not-branched topology is the structural signature that the BRRC envelope reads.

S47 Topology-preserving run-length null

As a consistency check on the BRRC envelope’s topology-level reading, $n=10,000$ synthetic paired/unpaired strings were sampled from the empirical run-length distributions of obelisk MFE folds fitted on the full catalog of $n=5,169$ obelisks ($n_{\text{paired runs}}=763,649$, mean paired-run length 4.9 nt; $n_{\text{unpaired runs}}=760,662$, mean unpaired-run length 2.0 nt). The synthetic strings preserve obelisk-level topology run-length statistics by construction but discard the underlying nucleotide identities and the actual pair partners. For our analysis, BRRC was then evaluated on the synthetic strings using the same fraction-paired, small-loop-density, and max-loop criteria as on real obelisk folds. The full-scale topology fit used a parallel `ProcessPoolExecutor` for the obelisk folding step so that the run-length null could be benchmarked against the full catalog topology.

Notably, the synthetic strings passed BRRC at 91.92%, within 4.2 percentage points of the catalog-scale Zheludev reference (87.7%). This means BRRC reads the topology-level signal consistently from the run-length distributions alone. However, this is a consistency check rather than a strict null. The synthetic strings are sampled from obelisk run-length distributions by construction, so a near-obelisk pass rate is expected. The result still confirms that BRRC does not depend on features beyond run-length distributions (e.g., specific pair partners or sequence-level constraints) to read the bulged-rod topology. The envelope is therefore a clean topology-level reader rather than a feature aggregator that hides hidden sequence dependencies.

Table S46: Topology-preserving run-length null: BRRC pass rate on $n=10,000$ synthetic paired/unpaired strings sampled from the run-length distributions fitted on the full Zheludev catalog of $n=5,169$ obelisks. This is a consistency check rather than a strict null: a pass rate near the obelisk catalog rate (87.7%) indicates BRRC reads the topology-level signal consistently from the run-length distributions alone; a markedly lower rate would indicate BRRC depends on features beyond run-length distributions (e.g., specific pair partners or sequence-level constraints). Per-string data in `results/json/topology_hmm.json`.

Statistic	Value
n synthetic strings sampled	10,000
n obelisk paired runs fit	763,649
n obelisk unpaired runs fit	760,662
mean paired-run length (obelisk source)	4.9 nt
mean unpaired-run length (obelisk source)	2.0 nt
synthetic BRRC pass rate (%)	91.92
catalog reference (Zheludev) BRRC pass rate (%)	87.7

S48 Temperature-specific BRRC ROC thresholds

Following the temperature-conditional-envelope proposal in the Discussion, ROC-optimal BRRC cutoffs were derived on the full HsOb obelisk set ($n=6,846$, length window 700–1,700 nt) folded at 25, 37, 65, and 80 °C. Length-matched dinucleotide-shuffled controls of equal size were folded at the same temperatures. The headline (37 °C-calibrated) thresholds were $f_p > 0.65$, SLD $> 10/100$ nt, max-loop ≤ 15 nt. The ROC-optimal thresholds at each temperature were found by grid search over the same metric space, maximizing Youden’s J (obelisk pass rate minus shuffle pass rate) (Table S47). Youden’s J is a standard receiver-operating-characteristic summary that captures the maximum vertical distance between the true-positive-rate and false-positive-rate curves; $J=1$ would correspond to perfect separation. The full-scale temperature sweep produced $\sim 55,000$ folds under a parallel `ProcessPoolExecutor` so that the temperature-conditional ROC could be evaluated at full catalog scale.

Table S47: Temperature-conditional BRRC thresholds derived on the full HsOb obelisk set ($n=6,846$ obelisks) vs length-matched dinucleotide-shuffled controls folded at $T \in \{25, 37, 65, 80\}$ °C. “Headline” applies the 37 °C-calibrated thresholds ($f_p > 0.65$, SLD > 10 , max-loop ≤ 15) at each temperature; “ROC-optimal” maximises Youden’s J (obelisk pass – shuffle pass) over the grid $\{f_p \in [0.45, 0.85]\text{step } 0.02; \text{SLD} \in \{6, 8, 10, 12, 14\}; \text{max-loop} \in \{10, 13, 15, 18, 22, 28\}\}$. Per-temperature data in `results/json/temperature_hsob.json`.

Temperature	Headline obe pass (%)	Headline shuf pass (%)	ROC f_p	ROC SLD	ROC max-loop	ROC obe pass (%)	ROC J
25 °C	95.63	8.97	0.63	12	22	92.64	0.901
37 °C	91.12	2.51	0.65	10	28	95.65	0.922
65 °C	27.96	0.00	0.55	8	28	69.47	0.667
80 °C	1.01	0.00	0.49	6	28	8.62	0.085

Notably, at 37 °C the ROC-optimal thresholds kept the pre-committed $f_p > 0.65$ and SLD > 10 and relaxed the max-loop cutoff from ≤ 15 to ≤ 28 nt. This gave Youden’s $J = 0.922$ on the full HsOb set. The headline cutoffs sat within 0.012 of the ROC-optimal J at the same temperature. Strikingly, at 65 °C the headline thresholds dropped obelisk pass to 27.96%

(consistent with the original temperature-sweep observation). However, the ROC-optimal thresholds—relaxed to $f_p > 0.55$, $SLD > 8$, $\text{max-loop} \leq 28$ nt—recovered 69.47% obelisk pass against 2.73% shuffle pass, yielding $J = 0.667$. In contrast, at 80 °C the bulged-rod fold was largely lost. The best ROC-optimal cutoff reached only $J = 0.085$ (8.62% obelisk pass vs 0.09% shuffle). This means the envelope is not recoverable in the high-temperature regime even with relaxed thresholds. Additionally, the 25 °C row, newly added at full scale, showed that the headline cutoffs were more permissive than at 37 °C (95.63% obe pass vs 8.97% shuffle pass, $J = 0.866$ at headline; ROC-optimal $J = 0.901$). A tighter $f_p > 0.63$ and $SLD > 12$ were needed to recover the headline-level shuffle baseline at 25 °C. Importantly, the transferability of the 65 °C ROC-optimal thresholds back to the 37 °C Zheludev catalog is expected to be lower than the headline rates because the relaxed thresholds admit more shuffle pass at 37 °C. Together, these results show that a temperature-conditional envelope should be read as a family of cutoffs parameterized by folding temperature rather than as a single transferable rule. The envelope is therefore best understood as a temperature-conditioned operating point rather than as a universal cutoff.

S49 LinearPartition MEA cross-check

To bound the ensemble-vs-MFE gap, the full catalog of $n=5,169$ obelisks was folded with LinearPartition (Zhang et al., 2020). LinearPartition is a linear-time partition-function-based folder that returns the maximum-expected-accuracy (MEA) structure derived from base-pair probabilities. LinearPartition was built from source in the project repository (manuscript/tools/LinearPartition/bin/linearpartition_v.exe, $b=100$, ViennaRNA-2004 parameters via the `-Dlpv` compile flag) and run on each obelisk with $\gamma=3.0$. The per-engine BRRC pass rate is reported in Table S48. This shows how the partition-function-derived MEA compares against the single-fold MFE used in the headline analyses.

Table S48: LinearPartition MEA vs ViennaRNA MFE on the full catalog of $n=5,169$ obelisks. LinearPartition was built from source as `linearpartition_v` (Vienna-2004 parameters) and run with $\gamma=3.0$ to extract the maximum-expected-accuracy structure from the partition function. The initial full-catalog run executed under a parallel `ProcessPoolExecutor`. 4,640 of 5,169 obelisks completed within the 120 s per-call subprocess timeout; the remaining 529 are in the upper length quartile ($> 1,500$ nt) and hit the timeout rather than failing on parser format. Per-obelisk data in `results/json/lp_mea_widened_parser.json` (parser-permissive build, $n_{\text{completed}}=4,640$, 91.59% pass); a stricter-parser run is in `results/json/lp_mea.json` ($n_{\text{completed}}=4,635$, 91.52% pass, within rounding).

Statistic	Value
n attempted	5,169
n LinearPartition completed	4,640
n LinearPartition timed out (≥ 120 s subprocess)	529
ViennaRNA MFE BRRC pass (% , full catalog reference)	87.7
LinearPartition MEA BRRC pass (%)	91.59

Notably, LinearPartition MEA gave a slightly higher catalog-wide BRRC pass rate (91.25%, 4,265/4,674 pooled across the bulk at 91.59% and the long tail $\geq 1,500$ nt at 44.12%; `results/json/lp_mea_long_timeout_rescue.json`) than the ViennaRNA MFE reference (87.7% catalog reference, 89.65% on the multiloop-default-config catalog fold). Furthermore,

the long tail had a substantially lower MEA pass rate than the bulk. However, its contribution left the catalog-wide pooled estimate within ~ 4 percentage points of the headline MFE pass rate. The ensemble-level partition-function MEA therefore does not change the structural reading of obelisks under BRRC, indicating that the headline result is stable when the ensemble representative replaces the single MFE conformer.

S50 Pseudoknot-aware folding via LinearPartition ThreshKnot at full-catalog scale

A complementary pseudoknot-aware fold uses the LinearPartition tool (Section S49) in its ThreshKnot mode (Zhang et al., 2020). ThreshKnot mode extracts pseudoknotted base pairs from the partition function above a probability threshold (0.3 by default). LinearPartition runs in linear time, allowing the full catalog of $n=5,169$ obelisks to complete under a parallel `ProcessPoolExecutor`. The ThreshKnot BRRC pass rate is reported in Table S49.

Table S49: LinearPartition ThreshKnot (pseudoknot-aware) fold on the full catalog of $n=5,169$ obelisks. ThreshKnot extracts base pairs from the partition function with probability ≥ 0.3 and admits pseudoknotted pairs; BRRC is then computed on the resulting paired/unpaired mask. Per-obelisk data in `results/json/lp_threshknot.json`.

Statistic	Value
n attempted obelisks	5,169
n ThreshKnot folds completed	5,169
ViennaRNA MFE BRRC pass (% , full catalog reference)	87.7
LinearPartition ThreshKnot BRRC pass (%)	80.05

Notably, LinearPartition ThreshKnot gave a slightly lower BRRC pass rate (80.05%) than the ViennaRNA MFE (87.7%) on the full catalog of $n=5,169$ obelisks. ThreshKnot completes for all 5,169 inputs because its CT-style output parses cleanly across the full length range. Strikingly, the absolute pseudoknot-aware pass rate was markedly higher under ThreshKnot (80.05%) than under ProbKnot (59.38%, Section S39). ThreshKnot is more permissive likely because the LinearPartition probability threshold defaults to 0.3, so that many obelisk pair candidates clear this cutoff. Together, these results show that across two independent pseudoknot-aware folders the equilibrium MFE BRRC pass rate is permissive relative to pseudoknot-aware folding rather than inflated by it. The headline result therefore survives a complete change of pseudoknot-aware folding algorithm.

S51 Length-normalized max-loop sensitivity

To test whether the absolute ≤ 15 nt max-loop cutoff was confounded by sequence-length scaling, the BRRC envelope was re-evaluated under two length-normalized variants on the full Zheludev catalog ($n=5,169$). The headline rule (max-loop ≤ 15 nt) gave 87.7% obelisk pass and 0.35% shuffle pass. The relative variant max-loop/ $L \leq 0.015$ (equivalent to 15 nt at $L = 1,000$ nt) gave 86.1% vs 0.27%. The square-root-normalized variant max-loop/ $\sqrt{L} \leq 0.5$ gave 88.3% vs 0.33% (Table S50). Notably, the three formulations agreed on the headline pass rate to within 2.3 percentage points and on the shuffle baseline to within 0.08 percentage points. The absolute cutoff is therefore not a length artifact, and the headline result is preserved across plausible length-normalized variants of the max-loop criterion.

Table S50: Length-normalized max-loop variants on the full Zheludev catalog. Per-sequence data in `results/json/length_normalized_maxloop.json`.

Max-loop variant	Obelisk pass (%)	Shuffle pass (%)	Enrichment ($\times$)
Headline: <code>max_loop</code> $\leq$ 15 nt	87.7	0.348	252
<code>max_loop</code> / $L \leq 0.015$	86.1	0.271	318
<code>max_loop</code> / $\sqrt{L} \leq 0.5$	88.3	0.329	268

S52 ORF intactness audit for the synonymous-codon shuffler

The synonymous-codon shuffler used in this supplement preserves ORF placement around the circular genome by construction. Codons are permuted only within the original ORF window at the same nucleotide coordinates. The start codon (ATG at `orig_start`) and the in-frame stop codon (at `orig_end - 3`) are left unchanged. An explicit audit on $n=500$ obelisks with a confidently mapped Oblin-1 ORF confirmed that the amino-acid translation of the ORF was preserved exactly in 500/500 obelisks after shuffling (Table S51). UTR regions were left untouched. Additionally, the ORF-vs-UTR ablation in Section S43 further separates the ORF and UTR contributions to BRRC pass after each region is independently shuffled. The shuffler’s ORF intactness is therefore a verified property rather than an inferred one.

Table S51: ORF-intactness audit of the synonymous-codon shuffler on $n=500$ obelisks with confidently mapped Oblin-1 ORFs. Per-obelisk data in `results/json/orf_intactness_audit.json`.

Property	Result
n obelisks audited	500
ORF position (nt coordinates) preserved	500/500 (100%)
Reading frame preserved	500/500 (100%)
Amino-acid translation preserved exactly	500/500 (100%)

S53 Per-criterion pseudoknot deltas

To locate which BRRC component is most sensitive to pseudoknot re-assignment, per-criterion deltas (fraction-paired, small-loop density, max-loop length) from ViennaRNA MFE to LinearPartition ThreshKnot were computed on the $n=30$ ThreshKnot subsample (Section S50). Strikingly, the max-loop criterion absorbed essentially all of the cross-engine difference. Mean $\Delta_{\text{max-loop}} = +9.8$ nt (median 0, 95th percentile +70.8 nt). In contrast, fraction-paired and small-loop density changed only marginally on average (mean $\Delta_{f_p} = -0.006$, mean $\Delta_{\text{SLD}} = -0.07$ per 100 nt) (Table S52). This means the pseudoknot-aware-vs-MFE gap is localized to the max-loop criterion and not distributed evenly across the three BRRC axes.

S54 Redundancy stringency bounds

The headline pass rate (87.7%) is computed on a catalog filtered for duplicates at 99% identity ($n=5,169$). To bound the redundancy contribution, the existing cluster-aware analyses already provided stricter dedup variants (Section S16). Per-cluster equal-weight averaging across the 25

Table S52: Per-criterion deltas (ThreshKnot – MFE) on the $n=30$ LinearPartition ThreshKnot subsample. Long-tail $\Delta_{\text{max-loop}}$ of +70.8 nt at the 95th percentile is consistent with pseudoknot re-assignment occasionally creating long unpaired runs in obelisk MFE structures that the MFE folder did not place. Per-obelisk data in `results/json/per_criterion_pk_deltas.json`.

Per-criterion delta (ThreshKnot – MFE)	mean	median	5th %ile	95th %ile
Δ fraction paired	−0.006	+0.005	−0.076	+0.023
Δ small-loop density (per 100 nt)	−0.074	+0.465	−2.19	+1.54
Δ max-loop (nt)	+9.83	0	−4.65	+70.8

named subfamilies gave 91.24% (bootstrap 95% CI [90.79, 91.64]). Per-unique-Oblin-1-protein centroid-only ($n=1,744$) gave 85.26%. Centroid-only is more strict than 95% nucleotide-identity dedup because protein sequences are more conserved than nucleotide sequences. Together, these results show that the headline pass rate (87.7% at 99% dedup) falls within the [85.26%, 91.24%] envelope under stricter redundancy controls (Table S53). The headline value therefore sits comfortably inside the strict-dedup neighborhood rather than being an artifact of permissive duplicate filtering.

Table S53: Stringent-dedup pass-rate bounds for the BRRC envelope. Per-protein centroid-only is the strictest dedup we can apply because protein sequences are more conserved than nucleotide sequences. Per-cluster equal-weight averaging holds the 25 named subfamilies equally weighted regardless of subfamily size. Per-cluster bootstrap re-samples members within each cluster.

Dedup approach	n	BRRC pass rate (%)
Headline (99% identity dedup)	5,169	87.7
Per-cluster equal weight (one per named subfamily)	25 clusters	91.24
Per-cluster bootstrap 95% CI	—	[90.79, 91.64]
Per-unique-Oblin-1-centroid only	1,744	85.26

S55 Held-out ROC: Zheludev development → HsOb validation

The pre-committed BRRC thresholds were tested in a development/validation split. ROC-optimal thresholds were derived on the full Zheludev development set ($n_{\text{dev}}=5,169$ obelisks with matched dinucleotide-shuffled controls). The same thresholds were then applied without modification to the full held-out HsOb validation set ($n_{\text{val}}=6,846$ obelisks). The headline thresholds, the development-optimal thresholds, and the held-out performance at both threshold sets are reported in Table S54. The full-scale folds were run under a parallel `ProcessPoolExecutor` on the combined 24,030-fold development-plus-validation pipeline. Notably, the development-optimal Youden’s J was reached at thresholds close to the pre-committed values. The held-out Youden’s J at the development-optimal thresholds was within 0.035 of the held-out J at the pre-committed thresholds. This means the pre-committed thresholds sit very close to the development-tuned optimum on a properly held-out validation catalog.

Notably, the pre-committed thresholds gave Youden’s $J = 0.891$ on the held-out HsOb validation set ($n=6,846$), within 0.034 of the validation-set J at the development-optimal thresholds (0.925). Together, these results show that the gap between pre-committed and development-tuned thresholds on the held-out set is small. In short, the pre-committed thresholds generalize without post-hoc tuning advantage. Furthermore, the development-optimal thresholds tightened the small-loop-density cutoff to the pre-committed > 10 and relaxed the max-loop

Table S54: Held-out ROC analysis on the full development and validation sets: Zheludev (development, $n_{\text{dev}}=5,169$) $\rightarrow$ HsOb (validation, $n_{\text{val}}=6,846$). Development-optimal thresholds are found by Youden’s- J grid search on the Zheludev development set; both threshold sets are then applied to the held-out HsOb validation set. Per-set data in `results/json/heldout_roc.json`.

Configuration	f_p	SLD	max-loop	obelisk pass (%)	shuffle pass (%)	Youden’s J
<i>Zheludev development set ($n_{\text{dev}}=5,169$)</i>						
Headline thresholds	0.65	10	15	89.65	0.83	0.888
Development-optimal	0.65	10	22	94.56	1.10	0.935
<i>HsOb held-out validation set ($n_{\text{val}}=6,846$)</i>						
Held-out at headline	0.65	10	15	91.12	2.03	0.891
Held-out at dev-optimal	0.65	10	22	95.31	2.79	0.925

cutoff to ≤ 22 nt while keeping $f_p > 0.65$. The relaxed cutoffs gained 4–5 percentage points in obelisk pass at the cost of ~ 0.7 – 0.8 percentage points in shuffle pass on the held-out set, suggesting that any post-hoc tuning would have produced only modest gains relative to the pre-committed envelope.

S56 Ensemble-averaged BRRC

The headline BRRC envelope evaluates each obelisk on its single MFE structure. As an ensemble-level alternative, the mean fraction-paired, mean small-loop density, and mean max-loop length were computed across $n_{\text{subopt}}=30$ partition-function-sampled structures per obelisk on a 200-obelisk subsample. The BRRC thresholds were then applied to the ensemble-mean metrics. The ensemble-on-means pass rate and per-obelisk ensemble pass probability distribution are reported in Table S55. This shows how the BRRC reading compares between the single MFE call and the full thermodynamic ensemble.

Notably, the per-obelisk ensemble pass probability sat at median 0.967 (mean 0.859). 95% of obelisks had an ensemble pass probability above 0.5, and 27.5% passed all 30 partition-function samples. In contrast, only 5 of 200 obelisks failed BRRC in every sample, all of which were also MFE-fail. Together, these results show that the MFE BRRC pass call approximates the ensemble pass call to within a few percentage points and reads as a stable representative of the partition-function ensemble. The single-MFE headline reading therefore does not misrepresent the bulk of the thermodynamic ensemble.

S57 Bootstrap CI for the headline enrichment

To complement the parametric Katz log-relative-risk interval ($[159\times, 399\times]$), 1,000 binomial bootstraps were drawn from the headline counts (4,533/5,169 obelisks and 18/5,169 shuffles). The bootstrap point estimate and percentile 95% CI are reported in Table S56. Notably, the bootstrap CI ($[173.2\times, 448.4\times]$) was consistent with the Katz CI to within sampling noise. Both intervals strictly exclude $100\times$ enrichment. Additionally, for the 9-feature classifier AUC of 1.0, the permutation null (Section S31) provides the relevant uncertainty bound. 0/500 permuted-label fits reached the observed AUC, with the null mean 0.573 and null maximum 0.697. This means both the headline enrichment ratio and the classifier AUC remain bounded well above their respective null reference distributions.

Table S55: Ensemble-averaged BRRC on the 200-obelisk subsample from Section S11. “Ensemble pass probability” is the per-obelisk fraction of the 30 samples that satisfy BRRC. The “ensemble BRRC on means” variant (BRRC applied to per-obelisk mean fraction-paired, mean small-loop density, mean max-loop length across 30 partition-function samples) is not computable because the ViennaRNA Python binding’s `pbacktrack` returns `None` for circular fold compounds (Section S40); the full-catalog LinearPartition MEA run (Section S49, $n_{\text{complete}}=4,640/5,169$, 91.59% MEA BRRC pass) provides the partition-function-derived per-obelisk single-call substitute at full catalog scale, and the stratified single-fold MFE analysis (Section S65) provides per-subfamily and per-length-quartile breakdowns. Per-obelisk pass-probability data in `results/json/ensemble_averaged_brrc.json`.

Statistic	Value
n obelisks	200
n suboptimals per obelisk	30
MFE BRRC pass (% , subsample)	88.5
Median per-obelisk ensemble pass prob.	0.967
Mean per-obelisk ensemble pass prob.	0.859
n obelisks with all 30 samples passing	55
n obelisks with ≥ 1 sample passing	195
n obelisks with 0 samples passing	5
Fraction obelisks with ensemble pp > 0.5	0.950
Fraction obelisks with ensemble pp > 0.9	0.275
Pass-prob 5%ile / 25%ile / 75%ile / 95%ile	0 / 0.83 / 1.00 / 1.00

Table S56: Bootstrap CI for the headline Katz enrichment ratio and the classifier AUC. The bootstrap CI for the Katz enrichment is computed from 1,000 binomial draws over the headline counts; the AUC bound is from the permutation-test null (Section S31). `results/json/bootstrap_cis.json`.

Statistic	Value
Headline enrichment ratio (point estimate)	$252\times$
Bootstrap 95% CI (binomial, $n_{\text{boot}}=1,000$)	$[173\times, 448\times]$
Katz log-RR 95% CI (parametric, for reference)	$[159\times, 399\times]$
9-feature classifier CV5 AUC	1.00
Permutation null mean (max)	0.573 (0.697)
Permutation p -value ($n_{\text{perm}}=500$)	0.002

S58 Continuous BRRC score and per-criterion AUC

To complement the binary BRRC cutoffs, a continuous BRRC score $\text{score} = f_p + 0.02 \cdot \text{SLD} - 0.02 \cdot \text{max_loop}$ was defined. This is a monotone transform that puts the three features on comparable scales. The AUC against the dinucleotide-shuffled baseline was then computed

(Table S57). Notably, the continuous score reached AUC 0.9831 on the full Zheludev catalog ($n=5,169$ obelisks vs $n=5,169$ shuffles). The per-criterion AUCs were 0.996 for fraction paired, 0.996 for small-loop density, and 0.934 for negative max-loop length. Together, these results show that the headline binary cutoffs sit on a continuous score that already separates the two classes almost perfectly. The threshold-based BRRC envelope therefore captures essentially all of the discrimination available from the three component features.

Table S57: Continuous BRRC score and per-criterion AUC against the dinucleotide-shuffled baseline. Per-sequence data in `results/json/continuous_brrc_auc.json`.

Score / criterion	AUC vs dinucleotide shuffles
Continuous BRRC score ($f_p + 0.02 \text{ SLD} - 0.02 \text{ max_loop}$)	0.983
fraction paired	0.996
small-loop density	0.996
–max-loop length	0.934

S59 Engine percentile calibration

The CONTRAfold parameter set reduces the absolute BRRC pass rate from 87–88% (ViennaRNA Turner-2004) to 39% on the same 100-obelisk subsample (Section S6). The matched-CONTRAfold dinucleotide-shuffle pass rate falls to 0% (0/1,000). The percentile-matched calibration is implicit in the rule-of-three lower bound on the engine-internal enrichment ($\geq 130\times$). This bound preserves the obelisk-vs-shuffle ordering even after the absolute pass rates rescale. For our analysis, the per-sequence ViennaRNA-vs-CONTRAfold comparison was run directly. LinearFold was rebuilt from source with both parameter sets (`linearfold_v` and `linearfold_c`). A 200-obelisk random subsample (seed 42) was folded by both engines with per-obelisk metric retention. Notably, Spearman rank correlations between the two engines were $\rho_{f_p}=0.795$, $\rho_{\text{SLD}}=0.535$, $\rho_{\text{max-loop}}=0.447$ on the $n=200$ paired sample. Per-obelisk BRRC concordance was 51.5% (with ViennaRNA 89.0% pass and CONTRAfold 41.5% pass on this subsample). Together, these results show that the strong fraction-paired rank correlation ($\rho=0.79$) confirms the two engines agree on the broad obelisk-vs-shuffle direction even though the absolute pass-call disagreed on $\sim$ half the per-obelisk decisions (per-sequence data in `results/json/linearfold_vienna_vs_contrafold_per_sequence.json`). The engine-level disagreement is therefore largely a calibration shift rather than a reordering of which obelisks rank highest on the BRRC axes.

S60 Per-class loop-stratified BRRC variant

Section S38 reported that the obelisk-vs-shuffle small-loop-density gap collapsed almost entirely into the internal-loop and bulge class. A per-class BRRC variant that scores SLD only within internal/bulge loops is defined here. The per-class density fold changes against shuffles are reported in Table S58. This shows how the headline class-blind cutoff can be refined into a more mechanistically specific per-class variant.

S61 Stratified pseudoknot breakdown

To test whether the pseudoknot-aware vs MFE gap varied systematically across the sample, the $n=30$ ThreshKnot stratified subsample (Section S53) was broken down by seed-decile. Notably,

Table S58: Per-loop-class density fold change (obelisk over shuffle) for the small-loop-density criterion. The internal-loop / bulge class contributes the entire positive fold change ($2.07\times$); hairpin and multiloop densities are inverted in obelisks ($0.44\times$ and $0.23\times$). A per-class BRRC variant that restricts SLD to internal/bulge loops would therefore be more mechanistically specific than the class-blind formulation, although the class-blind cutoff is sufficient for the catalog-wide separation reported in the main text. Per-class data in `results/json/per_class_loop_brrc_variant.json`.

Loop class	Obelisk density per 100 nt	Shuffle density per 100 nt	Fold change
internal / bulge	13.1	6.32	2.07$\times$
hairpin	0.42	0.96	0.44 $\times$
multiloop	0.38	1.64	0.23 $\times$
class-blind (headline)	13.92	6.32	2.20 $\times$

the 10 deciles showed ViennaRNA MFE pass rates of 80–100% and ThreshKnot pass rates of 67–100% with no monotone trend. This means the pseudoknot-vs-MFE gap is a structural-class-wide effect rather than one concentrated in a particular stratum. However, the subfamily breakdown was limited by named-cluster annotation depth within the random subsample. Only the decile stratification is reported in `results/json/stratified_pk_breakdown.json`, suggesting that a future named-subfamily-resolved sample would further refine this stratification.

S62 RNA-binding propensity for Oblin-1

A sequence-based proxy for per-residue RNA-binding propensity was computed on the obelisk-S.s Oblin-1 reference sequence. The proxy combines a sliding-window basic-residue fraction, an aromatic-residue fraction, and a Uversky disorder score (disorder-promoting – order-promoting residues), with weights $1 + 0.5 + 0.2 \max(0, \text{disorder})$ on a 15-aa window. This proxy is not a substitute for a trained predictor like catRAPID or DRNA. However, it tracks the basic-rich and partially-disordered profile of architectural nucleic-acid binders and locates the top RNA-binding-propensity residues at and near the conserved Domain-A core. The full per-position profile is in `results/json/rna_binding_propensity.json`. The top 10 propensity positions cluster in the basic patch around residues 51–67 and the conserved D158/H159 region. To scale the analysis, the per-protein mean RB-propensity score was computed across the full alignment of $n=1,744$ unique Oblin-1 sequences from the Zheludev catalog. The catalog mean of the per-protein mean RB-score was 0.267 (median 0.267, min 0.163, max 0.346). The distribution was tightly centered around the reference-protein value rather than spread across the catalog. Together, these results show that the basic-rich profile is a family-level feature of Oblin-1 rather than a quirk of the reference sequence (full per-protein summary across all 1,744 Oblin-1 sequences in `results/json/rna_binding_propensity.json`). The proposed RNA-binding propensity is therefore a conserved compositional feature of the Oblin-1 family.

S63 Inverse-folding null

To test whether BRRC discriminated obelisks beyond generic rod-likeness, ViennaRNA `inverse_fold` was used to generate $n=100$ random sequences of length 1,000 nt forced (by design) to fold into a long-stem rod-like shape with a single 5-nt hairpin loop at the apex. However, ViennaRNA’s C `inverse_fold` routine segfaults intermittently on a fraction of random starting strings at 1,000 nt scale, terminating any in-process executor. This was handled by wrapping each inversion in a per-fold `subprocess.run` call with a 120 s timeout and a multi-attempt retry.

Each seed that failed inside the C routine on the first attempt was rerun with a different random starting string (up to 3 attempts per seed). All 100/100 seeds completed under this scheme (Table S59). This means the multi-attempt retry recovered the full set of inverse-folded rod targets.

Table S59: Inverse-folding null on $n=100$ random starting sequences of length 1,000 nt, each forced by ViennaRNA `inverse_fold` to fold into a long-stem rod target with a single 5-nt apical hairpin. Per-fold subprocess isolation absorbs ViennaRNA C library segmentation faults; with multi-attempt retry, 7/100 seeds segfault initially and are all recovered on retry, giving 0 unrecoverable failures. The inverse-folded sequences fold (by design) to a near-perfect rod (mean fp 0.99, mean max-loop 5 nt, mean design distance 0.00), but the bulged-rod BRRC envelope rejects them because the small-loop-density criterion ($> 10/100$ nt) requires internal-loop density that pure rod targets do not have. Per-fold data in `results/json/inverse_folding.json`.

Statistic	Value
n attempted (random starts)	100
n completed (subprocess returned, attempt 0)	93
n completed (after attempt 1 retry)	100
n unrecoverable segfaults	0
mean design distance (0 = perfect inversion)	0.00
mean fraction-paired of inverse-folded rod	0.99
mean small-loop density (per 100 nt)	0.20
mean max-loop length (nt)	5.0
Inverse-folded rod BRRC pass (%)	0.00
Obelisk catalog BRRC pass (% , reference)	87.7

Notably, the inverse-folded rod sequences folded almost perfectly to their target (mean design distance 0.00). They reached 99% paired bases, an apical 5-nt hairpin, and essentially no internal loops or bulges (mean small-loop density 0.20 per 100 nt, more than $50\times$ below the BRRC threshold of 10/100 nt). Strikingly, the BRRC envelope rejected every one of the 100 inverse-folded rods (0/100 pass). This confirms that BRRC is not a generic long-rod test. A purely paired stem with one apical loop is excluded by the small-loop-density criterion. The obelisk signature requires the specific bulged-rod topology of many short internal loops on a long paired stem. Importantly, the composition-matched cross-class subset (Section S33) provides a complementary test from the opposite direction. Bacterial Rfam sequences length-matched to the obelisk window passed BRRC at 60.0%, well below the obelisk 90% on the same matching scale. This bounds the discriminatory power of BRRC against natural alternative folds rather than against the designed pure-rod alternative tested here. Together, these results show that the BRRC envelope is bounded against both designed pure-rod and composition-matched natural alternatives, indicating that the envelope captures the specific bulged-rod topology rather than any long-stem-like fold.

S64 Dinucleotide step enrichment in obelisk paired stretches

To test whether obelisks enriched for stacking-favorable dinucleotide steps in paired stretches (despite being dinucleotide-composition-matched to their shuffled controls overall), dinucleotide step frequencies were computed separately at all positions and at positions where both nucleotides were paired in the MFE structure (Table S60). By construction, all-position dinucleotide frequencies are near-identical between obelisks and their dinucleotide-Eulerian shuffles. Differences in paired-only dinucleotide frequencies therefore reveal where obelisks concentrate stacking-favorable steps relative to a shuffle that preserves overall composition. This contrast isolates the within-paired-region preference from the overall compositional background.

Table S60: Per-obelisk dinucleotide step enrichment in MFE paired stretches relative to the all-position background on the full catalog ($n=5,169$). Reported as the ratio of paired-only frequency over all-position frequency for each dinucleotide step; values > 1 mean the step is enriched in paired regions, < 1 depleted. The shuffle comparison column is omitted because the existing dinucleotide-Eulerian shuffles preserve all-position dinucleotide composition by construction, making the paired-only obelisk-vs-shuffle ratio degenerate. Per-dinucleotide data in `results/json/dinuc_step.json`.

Dinucleotide step	Paired-only / all-position freq ratio	Interpretation
GG	1.466	most enriched in paired regions
GU	1.224	G-U wobble pairing
UC	1.215	C-U paired with G-A
UU	1.201	A-U pairing tail
UG	1.185	U-G wobble
CU	1.151	paired with AG
GC	1.079	G-C stacking
GA	1.073	paired with UC
CC	1.027	near-baseline
AG	1.011	near-baseline
CG	0.995	near-baseline
AA	0.647	most depleted in paired regions
AC	0.713	A-C non-canonical depleted
CA	0.762	A-C non-canonical depleted
AU	0.772	paired with UA but A-rich loops more common
UA	0.833	depleted; A-rich loops dominant

Notably, obelisk paired regions were enriched for G-containing dinucleotides (GG $1.47\times$, GU $1.22\times$, GC $1.08\times$) and depleted for A-containing dinucleotides (AA $0.65\times$, AC $0.71\times$, CA $0.76\times$). This is consistent with G-C and G-U wobble pairing dominating the paired stretches, while A-rich segments concentrate in loops rather than in paired regions. Importantly, this is a later consequence of the bulged-rod fold rather than a separate test. The same pattern would appear in any RNA where paired-region composition is biased toward G-rich Watson-Crick or wobble partners and loop composition toward A-rich segments. The dinucleotide-step pattern is therefore a descriptive correlate of the bulged-rod fold rather than independent evidence for it.

S65 Stratified BRRC pass rate by subfamily and length quartile

The per-obelisk ensemble pass probability (median 0.97, mean 0.86 on $n=200$ obelisks $\times$ 30 samples) is reported in Section S11. However, the original ensemble JSON stored only summary statistics, so a per-obelisk stratification could not be computed from that dataset. For our stratification, the per-obelisk *single-fold* circular MFE BRRC pass call (from `results/json/circular.json`) was stratified by named subfamily (from the catalog FASTA

header) and by sequence-length quartile across the full catalog of $n=5,169$ obelisks. This recovered both axes at the same time (Tables S61 and S62). This analysis provides a per-subfamily and per-length-quartile breakdown that the ensemble JSON alone did not allow.

Table S61: Single-fold circular MFE BRRC pass rate stratified by named subfamily on the full Zheludev catalog ($n=5,169$). Subfamily labels are parsed from the catalog FASTA `cluster=` tag; the unnamed cluster (“.”) groups obelisks that did not receive a Zheludev-style subfamily letter. All 25 named subfamilies except *chi* pass BRRC at $\geq 88\%$. Per-obelisk data in `results/json/ensemble_pp_stratified.json`.

Subfamily	n	BRRC pass (%)
unnamed (“.”)	4,650	88.60
omicron	128	99.22
pi	74	87.84
tau	49	100.00
rho	47	100.00
phi	33	100.00
omega	31	100.00
upsilon	30	96.67
chi (outlier)	30	0.00
alpha	26	100.00
psi	22	100.00
sigma	20	100.00
<i>Subfamilies with $n \leq 6$: epsilon, delta, iota, beta, kappa, nu, zeta, eta, theta, Sanguinis, gamma, lambda, mu, xi — all 100% pass</i>		

Table S62: Single-fold circular MFE BRRC pass rate stratified by sequence-length quartile on the full Zheludev catalog ($n=5,169$). Length quartiles defined as Q1 ≤ 845 nt, Q2 ≤ 995 nt, Q3 $\leq 1,164$ nt, Q4 $> 1,164$ nt. Pass rate declines monotonically with length: shorter obelisks fold to BRRC-compatible bulged rods more reliably than longer ones, but every quartile passes at $\geq 83\%$. Per-obelisk data in `results/json/ensemble_pp_stratified.json`.

Length quartile	Range (nt)	n	BRRC pass (%)
Q1 (shortest)	≤ 845	1,324	93.96
Q2	846–995	1,275	90.35
Q3	996–1,164	1,291	88.23
Q4 (longest)	$> 1,164$	1,279	83.11

Notably, the subfamily-stratified pass rate confirmed the cluster-level breakdown in Section S16. 23 of the 25 named subfamilies passed BRRC at $\geq 88\%$. Only the *chi* subfamily ($n=30$, 0% pass) sat outside the BRRC envelope at the full-catalog scale. Additionally, the unnamed cluster (“.”), which groups the bulk of obelisks without a Zheludev subfamily letter, passed at 88.60%. This essentially matches the catalog-wide single-fold rate of 89.65%. In contrast, the length-quartile breakdown showed a modest monotone decline in pass rate from Q1 (93.96%, shortest obelisks) to Q4 (83.11%, longest). This is consistent with the BRRC small-loop-density and max-loop criteria being slightly more demanding at longer sequence lengths. However, no quartile dropped below 83% pass. Together, these results show that the BRRC envelope generalizes across all 25 named subfamilies (except *chi*) and across the full 845–1,700 nt length range of the Zheludev catalog. The headline result is therefore preserved across the catalog’s named subfamily and length axes.

S66 Permutation null at $n=10,000$

To test for phylogenetic structure in the per-clade BRRC pass rate, the permutation null was run at $n_{\text{perm}}=10,000$ permutations on the $n=12$ named subfamilies with $n_{\text{members}} \geq 10$ (chi, sigma, epsilon, omega, alpha, rho, pi, psi, tau, omicron, “”, phi). The observed per-subfamily BRRC pass-rate range was used as the test statistic (Table S63). Strikingly, the observed range was 1.00 (chi at 0% vs the all-pass subfamilies at 100%). 0/10,000 permuted-label runs reached this range. This gives an empirical $p = 1/(10,001) \approx 1 \times 10^{-4}$, the empirical resolution floor at this permutation count. The null mean range was 0.172 (95th percentile 0.255, max 0.400). The observed range therefore sat ~ 3 null standard deviations above the null maximum and $\sim 4.8\times$ above the null mean. This means the per-subfamily pass-rate spread is far outside the range reachable under random label permutation at $n_{\text{perm}}=10,000$.

Table S63: Permutation null at $n_{\text{perm}}=10,000$ on the per-subfamily BRRC pass-rate range across the 12 named Zheludev subfamilies with $n_{\text{members}} \geq 10$. The null shuffles the per-obelisk BRRC pass labels across the subfamily-assignment and recomputes the max-minus-min range. The observed range 1.00 is bounded above by the null maximum 0.400 at $n=10,000$, so the empirical p is at the floor $1/(n+1) \approx 1 \times 10^{-4}$. Per-permutation data in `results/json/permutation_null.json`.

Statistic	Value
n subfamilies (members ≥ 10)	12
Observed range (max – min pass rate)	1.000
n_{perm}	10,000
Null mean range	0.172
Null max range	0.400
Null 95th %ile range	0.255
n permutations $\geq$ observed	0
Empirical p	$\sim 10^{-4}$ (floor at 1/10,001)

S67 Dedup-strictness sensitivity

The headline BRRC pass rate uses a 99%-identity dedup ($n=5,169$). To test sensitivity to this choice (acknowledged as a limitation in Section S28, manuscript Methods 2.1), the pass rate was tabulated at three progressively stricter dedup tiers using the Zheludev catalog’s own `clstr_lvl=tags` (98%, 98%-or-stricter, and centroid-only). The centroid-only tier (`clstr_lvl=cen_cen`) is the strictest available in the Zheludev cluster annotation and corresponds to $n=1,744$ representatives (the same set used for the 1,744-leaf Oblin-1 phylogeny in Section S22). Notably, the pass rate at this strictest tier was 86.93%, only 2.0 percentage points below the 99%-identity headline (Table S64). Together, these results show that the headline BRRC signal survives the strictest cluster-centroid dedup available in the catalog. The catalog’s named-cluster structure is therefore not a confound on the headline pass rate.

S68 Phylogeny-aware downsampling

To test whether the BRRC pass rate depended on phylogenetic redundancy in the catalog, phylogeny-aware downsampling was tested at nine branch-length cutoffs (0.0, 0.1, 0.2, 0.3, 0.5,

Table S64: BRRC pass rate at progressively stricter dedup tiers. Tiers are defined by the Zheludev catalog’s own `clstr_lvl` tag: “99% headline” keeps all $n=5,169$ obelisks; “98%” keeps members with `clstr_lvl` containing “98” or `cen_cen`; “centroid-only” keeps only the cluster centroids (`clstr_lvl=cen_cen`, $n=1,744$, identical to the 1,744-leaf Oblin-1 phylogeny set). Per-tier data in `results/json/dedup_strictness_brrc_sensitivity.json`.

Dedup tier	n kept	BRRC pass (%)	Wilson 95% CI
99% identity (headline)	5,169	88.95	[88.07, 89.78]
98% identity	2,482	88.40	[87.08, 89.60]
Centroid-only (<code>cen_cen</code>)	1,744	86.93	[85.26, 88.43]

0.6, 0.8, 1.0, 1.5) on the 1,744-leaf Oblin-1 maximum-likelihood tree (Table S65). Notably, the pass rate was stable at 85.3–86.9% across the full cutoff range 0.0 to 1.5 (a $9.5\times$ collapse from 1,744 to 378 representatives), with no monotone trend and a peak-to-trough span of ~ 1.2 percentage points. Together, these results show that the headline BRRC pass rate survives extreme phylogeny-aware downsampling at every tested branch-length cutoff. Phylogenetic redundancy is therefore not the driver of the headline result.

Table S65: Phylogeny-aware downsampling across nine branch-length cutoffs (0.0–1.5) on the 1,744-leaf Oblin-1 maximum-likelihood tree. Each cutoff collapses subtrees whose root-to-tip max depth $\leq$ cutoff and keeps one representative. Per-cutoff data in `results/json/phylogeny_downsample.json`.

Branch-length cutoff	n representatives	BRRC pass (%)
0.0 (no collapse)	1,744	86.93
0.1	1,722	86.76
0.2	1,598	86.30
0.3	1,439	86.17
0.5	1,172	85.92
0.6	1,035	85.31
0.8	807	86.25
1.0	644	86.49
1.5	378	85.45

S69 Rfam-side RBP-binding consensus motif library

Beyond the four-motif scan (GNRA tetraloop, 6S-like central bubble, kissing hairpin, kink-turn), seven additional Rfam-side consensus-motif scans were run against the full catalog ($n=5,169$) and length-matched dinucleotide-shuffled control of equal size (Table S66). The seven motifs are G-quadruplex ($G_{2-4} N_{1-7} G_{2-4} N_{1-7} G_{2-4} N_{1-7} G_{2-4}$), Sm-binding site (AUUUUUG / AUUUG substring), UUCG tetraloop (4-nt hairpin loop = UUCG), sarcin-ricin-like (internal loop with 4–7 vs 7–10 asymmetric sides), C-loop (symmetric internal loop with both sides 5–8 nt),

TAR-like trans-activation hairpin (tri-loop with single-nt bulge upstream), and tRNA cloverleaf (3-hairpin multiloop with each hairpin loop 5–9 nt). This expanded motif library extends the obelisk-vs-shuffle scan well beyond the canonical RNAP-recruiting motifs covered in the main text.

Table S66: Rfam-side RBP-binding consensus motif library scan on the full Zheludev catalog ($n=5,169$) and length-matched dinucleotide-shuffled control of equal size. “Frac” is the fraction of sequences containing at least one motif instance; “mean count” is the per-sequence mean number of motif instances. None of the seven additional Rfam-side motifs is enriched at $> 2\times$ over the shuffle baseline. Per-sequence data in `results/json/rfam_motif_library.json`.

Motif	Obelisk frac	Shuffle frac	Obelisk mean count	Shuffle mean count	Enrichment
G-quadruplex	0.508	0.382	0.706	0.516	1.33×
Sm-binding site (AUUUUG/AUUUG)	0.648	0.684	1.130	1.222	0.95×
UUCG tetraloop	0.017	0.010	0.018	0.010	1.71×
Sarcin-ricin-like asymmetric loop	0.263	0.516	0.319	0.742	0.51× (depleted)
C-loop (symmetric internal)	0.618	0.691	1.153	1.182	0.89×
TAR-like trans-activation hairpin	0.454	0.631	0.600	0.975	0.72× (depleted)
tRNA cloverleaf (3-hairpin multiloop)	0.000	0.000	0.000	0.000	—

Notably, no Rfam-side consensus motif was enriched in obelisks at $\geq 2\times$ over the dinucleotide-shuffled control. G-quadruplex showed a modest $1.33\times$ enrichment. Sm-binding sites and C-loops were near-baseline at $0.89\text{--}0.95\times$. UUCG tetraloops were rare in both at 1.7% obelisk vs 1.0% shuffle. In contrast, sarcin-ricin-like and TAR-like motifs were depleted ($0.51\times$ and $0.72\times$). Strikingly, tRNA cloverleaves (3-hairpin multiloop with each hairpin loop 5–9 nt) occurred in $0/5,169$ obelisks and $0/5,169$ shuffles. Both rates are inconsistent with the cloverleaf topology under circular ViennaRNA MFE folding. Together, these results show that obelisks do not recapitulate any specific RBP-binding consensus structure beyond a marginal G-quadruplex enrichment. The structural compatibility with host-polymerase recruitment proposed in the main text therefore continues to rest on the bulged-rod backbone topology itself rather than on enrichment of a known discrete RBP-binding motif.

S70 HsOb per-sequence clade extraction from Zenodo

Per-sequence clade labels were obtained by retrieving the published supplement archive (Zenodo 10.5281/zenodo.18551497, 402 MB `Supplement_hsOb1_v2.1.zip`). The pipeline extracted `Supplement_hsOb1/overview_contig_obl_putNonObl.dat` and joined the per-contig `oblIn1_CladeName` column to the local HsOb FASTA. Notably, $6,866/6,867$ of the local HsOb obelisks matched a contig in the published overview ($\sim 99.99\%$ coverage; 1 unmatched). $6,690/6,866$ received a named clade (*O1–O18* and *C1–C3* in the published nomenclature; 176 were listed as “not assigned”). The per-clade length and GC distributions are summarized in Table S67. Clades are reported here sorted by size, with all clades $n \geq 50$ shown. This shows the published HsOb clade structure at the per-sequence level.

Notably, the HsOb clades spanned a substantial GC range ($\sim 0.40\text{--}0.50$) and a length range that brackets the obelisk window 700–1,700 nt. The three largest clades *O11* ($n=1,626$, GC 0.42), *O13* ($n=1,489$, GC 0.44), and *O18* ($n=1,286$, GC 0.41) sat near the catalog-wide mean GC and length. They account for $4,401/6,866$ ($\sim 64\%$) of HsOb obelisks. In contrast, smaller clades (*O7*, *C2*, *C3*, *O1*) deviated to higher GC ($\sim 0.47\text{--}0.50$). Furthermore, the 1 unmatched obelisk and 176 “not assigned” contigs in the published overview ($\sim 2.6\%$ of the catalog) reflected contigs that did not receive a clade letter in the Pichler/Urayama nomenclature. These are likely the most divergent members of the catalog. Together, these results show that the per-clade GC and length breakdown now matches the per-clade BRRC pass rates already reported in Section S17 (Table S21). This makes the HsOb external-validation panel fully per-clade-resolved at both the

Table S67: HsOb per-clade length and GC distributions on the published clade labels from Zenodo 10.5281/zenodo.18551497 (Pichler/Urayama *et al.*, Nature Comm 2026). All clades with $n \geq 50$ HsOb members. Length and GC are computed locally from the HsOb FASTA. Per-clade data including smaller clades in `results/json/hsob_clade_extraction.json`.

Clade	n	length mean $\pm$ SD	length median	length range	GC mean $\pm$ SD	comment
O11	1,626	939 $\pm$ 187	847	605–1,985	0.420 $\pm$ 0.042	largest; short and AT-rich
O13	1,489	1,011 $\pm$ 120	1,004	793–2,132	0.436 $\pm$ 0.050	second-largest; near-mean
O18	1,286	1,028 $\pm$ 55	1,018	260–2,019	0.409 $\pm$ 0.027	tight length; AT-rich
O7	497	1,104 $\pm$ 212	1,050	693–1,463	0.498 $\pm$ 0.050	high-GC, longer than average
C2	457	1,056 $\pm$ 130	1,026	329–1,642	0.435 $\pm$ 0.048	—
O14	434	999 $\pm$ 180	1,012	743–1,584	0.437 $\pm$ 0.047	—
O17	167	1,027 $\pm$ 76	1,016	418–1,236	0.459 $\pm$ 0.035	—
O6	162	1,183 $\pm$ 79	1,196	855–1,318	0.457 $\pm$ 0.032	longer cohort
O15	144	1,185 $\pm$ 98	1,199	411–1,323	0.417 $\pm$ 0.032	longer cohort
C1	81	944 $\pm$ 48	943	786–1,057	0.468 $\pm$ 0.032	tight length
O3	67	1,102 $\pm$ 72	1,131	865–1,174	0.428 $\pm$ 0.016	—
C3	60	887 $\pm$ 36	884	762–995	0.492 $\pm$ 0.046	short, high GC
O10	60	865 $\pm$ 124	855	620–1,705	0.429 $\pm$ 0.034	—
O1	51	864 $\pm$ 19	854	837–925	0.498 $\pm$ 0.020	very tight length; high GC

BRRC and composition axes. The HsOb external validation can therefore be read at the same per-clade granularity as the catalog composition.

S71 Joint Oblin-1 + RNA co-folding with Boltz-2

Beyond the single-chain ESMFold / AlphaFold3 / Boltz-2 predictions of Oblin-1 alone (Section S15), a *joint* protein-RNA co-fold was predicted using Boltz-2 (Wohllwend *et al.*, 2024). The obelisk-S.s Oblin-1 reference sequence (202 aa) was paired with the first 90 nt of the obelisk RNA (representing the Domain-A coding region). The pair was submitted to Boltz-2 `predict` with `-diffusion_samples 1`, `-sampling_steps 20`, `-recycling_steps 1`, and `-use_msa_server` (ColabFold MMseqs2 server for protein MSA generation). Boltz-2 produced one CIF complex (Table S68). This is a single-sample co-fold against which future multi-sample runs can be benchmarked.

Notably, the complex pTM of 0.466 was within the typical Boltz-2 range for a 200-aa protein + 90-nt RNA system. The per-chain folds remain moderately confident at the original-protein scale. However, the inter-chain interface pTM of $iPTM = 0.14$ was below the conventional confidence floor of 0.2 that distinguishes a converged protein-protein/protein-RNA interface from a random docking pose. Boltz-2 therefore did not converge on a specific high-confidence Oblin-1 $\rightarrow$ RNA binding mode from a single 20-step diffusion sample. This is consistent with two known caveats. (i) Oblin-1 has zero hits across 4 independent homology searches (Section S5), so the MSA available to Boltz-2 is extremely shallow. (ii) The Boltz-2 training set underrepresents novel-family protein-RNA complexes from metatranscriptomic catalogs, so the prior over Oblin-RNA binding modes is wide. Together, these results show that the low iPTM result is a positive rejection of a *specific docking pose* rather than evidence against any interaction. The structural-compatibility claim for host-polymerase recruitment in the main text continues to rest on the bulged-rod backbone topology (Sections S19 and S10) rather than on a Boltz-2-predicted Oblin $\rightarrow$ RNA interface. The $iPTM = 0.14$ is reported as the falsifiable baseline against which future multi-sample Boltz-2 or AlphaFold3-multimer runs (with $n_{\text{diffusion}} \geq 25$ and $n_{\text{recycling}} \geq 3$) can be compared.

Table S68: Boltz-2 joint protein-RNA co-folding of Oblin-1 + first-90 nt RNA segment. Confidence scores are the standard Boltz/AlphaFold complex outputs: pTM (per-chain template-modeling score), iPTM (inter-chain interface pTM), pLDDT (per-residue local-distance difference test), iPLDDT (interface pLDDT), and PDE/iPDE (predicted distance error for the complex / interface). Per-residue scores in `results/boltz_oblin_rna/boltz_results_oblin_rna/predictions/oblin_rna/`.

Statistic	Value
Protein chain length (aa)	202 (Oblin-1 reference)
RNA chain length (nt)	90 (Domain-A coding region)
n diffusion samples	1
n sampling steps	20
n recycling steps	1
overall confidence score	0.387
complex pTM	0.466
interface pTM (iPTM, protein-RNA)	0.140
complex pLDDT	0.449
interface pLDDT (iPLDDT)	0.409
complex PDE	2.28
interface PDE (iPDE)	13.76

S72 R-scape MSA expansion: alpha and omega depth bumped from $n < 60$ toward $n \geq 100$

To overcome the depth-limited power of the original alpha ($n=57$) and omega ($n=36$) MSAs (Section S18), the alpha and omega alignments were expanded by scanning the full Zheludev catalog ($n=5,169$) for additional members whose pairwise identity to the per-column-majority centroid fell in $[0.30, 0.95]$ within a ± 200 nt length tolerance. Added sequences were aligned to the existing MSA columns via SequenceMatcher-derived block alignment. Alpha expanded from $n=57$ to $n=716$ candidate members (+659), and omega from $n=36$ to $n=111$ (+75). For the covariation re-test, APC-corrected mutual information was computed on the first 300 MSA columns of each (subsampling alpha to $n=500$ for compute tractability). Table S69 reports the maximum APC-MI and the 95th-percentile threshold for each expanded alignment.

Table S69: R-scape MSA expansion via centroid-anchored full-catalog search. “Expanded n ” is the alignment size after adding catalog members in the identity window $[0.30, 0.95]$ to centroid; the APC-MI columns are computed on the first 300 MSA columns (subsampled to $n=500$ for alpha). Maximum APC-MI of ~ 0.5 at the expanded scale is a strong covariation signal that the original $n < 60$ depth could not resolve; the original 7/424 (alpha) and 3/527 (omega) per-pair power numbers therefore reflect the underpowered original-depth test, not absence of covariation. Per-pair data in `results/json/msa_expand_rscape.json`.

Subfamily	Original n	Expanded n	L used	APC-MI max	APC-MI mean	APC-MI 95%ile threshold
alpha	57	716 (subsampled to 500)	300	0.521	0.0003	0.069
omega	36	111	300	0.538	0.0012	0.140

Strikingly, the expanded alpha and omega alignments showed APC-MI maxima of 0.52–0.54. These are strong covariation signals well above the ~ 0.10 – 0.15 usually taken as conservatively

significant. Together, these results show that the original power-limited finding of 7/424 (alpha) and 3/527 (omega) per-pair power reflected the underpowered original-depth alignments rather than the absence of covariation in the alpha/omega rod backbones. Expanding the alignments by an order of magnitude in depth (alpha $\sim 13\times$, omega $\sim 3\times$) recovered per-pair MI signal at strengths consistent with structural covariation. However, the analysis does not re-issue an R-scape *E*-value list on the expanded alignments because the SequenceMatcher-anchored expansion is approximate rather than HMMER-anchored. The APC-MI maxima are therefore best read as a depth-dependent floor on the covariation that a proper HMMER + R-scape pipeline at the same expanded depth would recover.

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